

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396521
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Burns; Robert E. et al.

Tightening mechanisms and applications including the same

Abstract

This disclosure relates to articles that include a tightening mechanism, such as reel-based lace tightening mechanism, configured to tighten the article by rotation of a knob. The articles can include a concealing portion that is configured to conceal or protect at least a portion of the tightening mechanism, such as the knob. The concealing portion can be configured to prevent unintentional actuation of the tightening mechanism, such as during contact sports. The concealing portion can be configured to hide the tightening mechanism from view to improve the visual appearance of the article. The concealing portion can be collapsible such that a user can press the concealing portion down to expose the knob of the tightening mechanism.

Inventors: Burns; Robert E. (Denver, CO), Hammerslag; Gary R. (Steamboat Springs, CO), Irwin; Eric C. (Denver, CO), Lovett; Kristopher C. (Denver, CO), Nickel; Michael J. (Golden, CO), Soderberg; Mark S. (Conifer, CO)

Applicant: BOA Technology, Inc. (Denver, CO)

Family ID: 1000008777870

Assignee: BOA Technology, Inc. (Denver, CO)

Appl. No.: 18/739918

Filed: June 11, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240335008 A1	Oct. 10, 2024

Related U.S. Application Data

continuation parent-doc US 18316838 20230512 US 12042018 child-doc US 18739918
continuation parent-doc US 17142106 20210105 US 11684122 20230627 child-doc US 18316838
continuation parent-doc US 16126507 20180910 US 10888139 20210112 child-doc US 17142106

continuation parent-doc US 15231562 20160808 US 10070695 20180911 child-doc US 16126507
continuation parent-doc US 13829601 20130314 US 9375053 20160628 child-doc US 15191281
continuation parent-doc US 13098276 20110429 US 8516662 20130827 child-doc US 13973917
continuation-in-part parent-doc US 15191281 20160623 ABANDONED child-doc US 15231562
continuation-in-part parent-doc US 13973917 20130822 US 9408437 20160809 child-doc US
15231562
us-provisional-application US 61611418 20120315
us-provisional-application US 61330129 20100430

Publication Classification

Int. Cl.: **A43C11/16** (20060101); **A43B13/14** (20060101); **A43C1/04** (20060101); **A43C11/20** (20060101); **A61F5/01** (20060101)

U.S. Cl.:

CPC **A43C11/165** (20130101); **A43B13/14** (20130101); **A43C1/04** (20130101); **A43C11/20** (20130101); **A61F5/0118** (20130101);

Field of Classification Search

CPC: A43C (11/165); A43C (1/04); A43C (11/20); A43B (13/14); A61F (5/0118)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4633599	12/1986	Morell	242/389	A43C 11/165
7694354	12/2009	Philpott	2/416	A42B 3/08
7954204	12/2010	Hammerslag	36/50.5	A43C 11/16
8353087	12/2012	Chen	24/68SK	A43C 11/165
2006/0156517	12/2005	Hammerslag	24/68SK	A43C 11/00

Primary Examiner: Upchurch; David M

Attorney, Agent or Firm: Kilpatrick Townsend & Stockton LLP

Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/316,838 by Burns et al., entitled “TIGHTENING MECHANISMS AND APPLICATIONS INCLUDING THE SAME,” filed May 12, 2023, which is a continuation of U.S. Application Ser. No. 17/142,106 by Burns et al., entitled “TIGHTENING MECHANISMS AND APPLICATIONS INCLUDING THE SAME,” filed Jan. 5, 2021, now issued U.S. Pat. No. 11,684,122 on Jun. 27, 2023, which is a continuation of U.S. application Ser. No. 16/126,507 by Burns et al., entitled “TIGHTENING MECHANISMS AND APPLICATIONS INCLUDING THE SAME,” filed Sep. 10, 2018, now issued U.S. Pat. No. 10,888,139 on Jan. 12, 2021, which is a continuation of U.S. application Ser. No. 15/231,562 by Burns et al., entitled “TIGHTENING

MECHANISMS AND APPLICATIONS INCLUDING THE SAME,” filed Aug. 8, 2016, now issued U.S. Pat. No. 10,070,695 on Sep. 11, 2018, which is a continuation-in-part of U.S. application Ser. No. 15/191,281 by Burns et al., entitled “TIGHTENING MECHANISMS AND APPLICATIONS INCLUDING THE SAME,” filed Jun. 23, 2016, now abandoned, which is a continuation of U.S. application Ser. No. 13/829,601 by Burns et al., entitled “TIGHTENING MECHANISMS AND APPLICATIONS INCLUDING THE SAME,” filed Mar. 14, 2013, now issued U.S. Pat. No. 9,375,053 on Jun. 28, 2016, which claims priority to U.S. Provisional Application No. 61/611,418 by Burns et al., entitled “TIGHTENING MECHANISMS AND APPLICATIONS INCLUDING THE SAME,” filed Mar. 15, 2012. U.S. patent application Ser. No. 15/231,562 is also a continuation-in-part of U.S. application Ser. No. 13/973,917 by Goodman et al., entitled “REEL BASED LACING SYSTEM,” filed Aug. 22, 2013, now issued U.S. Pat. No. 9,408,437 on Aug. 9, 2016, which is a continuation of U.S. application Ser. No. 13/098,276 by Goodman et al., entitled “REEL BASED LACING SYSTEM,” filed Apr. 29, 2011, now issued U.S. Pat. No. 8,516,662 on Aug. 27, 2013, which claims priority to U.S. Provisional Patent Application No. 61/330,129 by Goodman et al., entitled “REEL BASED LACING SYSTEM,” filed Apr. 30, 2010. The entire disclosures of all aforementioned applications are hereby incorporated by reference, for all purposes, as if fully set forth herein.

BACKGROUND OF THE INVENTION

Field of the Disclosure

(1) Some embodiments of the present disclosure relate to articles (e.g., shoes, boots, braces, and other wearable articles) that use tightening systems (e.g., lacing systems).

Description of the Related Art

(2) Although various lacing systems are available for use in connection with various wearable articles, existing lacing systems suffer from various drawbacks. For example, some lacing systems include an exposed lace tightening mechanism, which can be visually unappealing. Also, during contact sports and some other uses, the exposed lace tightening mechanism can be damaged or unintentionally actuated (e.g., loosened). Accordingly, there persists a need for lacing systems that include a concealed or protected lace tightening mechanism.

BRIEF SUMMARY OF THE INVENTION

(3) Various embodiments disclosed herein relate to an article that includes a base material and a tightening mechanism coupled to the base material. The tightening mechanism can include a rotatable knob, and rotation of the knob in a tightening direction can tighten the article. The article can include a concealing portion that can extend upward from the base material and can at least partially radially surround the tightening mechanism. At least a portion of the rotatable knob can be rearward or inward of an outer surface of the concealing portion. In some embodiments, a majority of the rotatable knob can be rearward or inward of the outer surface of the concealing portion. In some embodiments, substantially the entire rotatable knob can be rearward or inward of the outer surface of the concealing portion. In some embodiments, a top surface of the rotatable knob can be substantially flush with the outer surface of the concealing portion.

(4) The concealing portion can include a compressible area, and compression of the compressible area can displace the outer surface of the concealing portion from a first position to a second position, and the second position can have a lower height than the first position. The compressible area can include compressible foam. The concealing portion can include a second foam material that is less compressible than the compressible foam, and the second foam material can at least partially radially surround the compressible foam. The compressible foam can be resilient and can facilitate return of the outer surface from the second position to the first position when a compressing force is not applied. The compressible area can include one or more collapsible recesses.

(5) The base material can include a hole, and at least a portion of the tightening mechanism can

extend through the hole in the base material.

(6) In some embodiments, the concealing portion can radially surround the tightening mechanism by a full 360 degrees.

(7) The concealing portion can include first and second areas on substantially opposite sides of the tightening mechanism from each other, and third and fourth areas on substantially opposite sides from each other. The heights of the first and second areas of the concealing portion can be greater than the heights of the third and fourth areas of the concealing portion such that the rotatable knob can be more exposed at the third and fourth areas than at the first and second areas.

(8) In one embodiment, an article (e.g., shoe, boot, apparel, and the like) may include a base material (e.g., heel, tongue, outsole, and the like) and a tightening mechanism coupled to the base material. The tightening mechanism may include a rotatable knob, wherein rotation of the knob in a tightening direction tightens the article. A compressible material may be coupled with a body (e.g., a housing) of the tightening mechanism. The compressible material may be positioned under a top layer of the base material so as to provide a transition between the body of the tightening mechanism and the base material to conceal edges of the body from view of a user. A concealing portion may extend upward from the base material and at least partially radially surround the tightening mechanism. At least a portion of the rotatable knob may be positioned rearward of an outer surface of the concealing portion so as to conceal the portion of the knob or the entire knob.

(9) In one embodiment, the compressible material may include a foam material having a durometer of between about 10 and about 25 Shore A. In some embodiments, a relatively rigid mounting component (e.g., a bayonet) may be coupled with the compressible material and the base material. The body of the tightening mechanism may be coupled with the mounting component to limit distortion of the compressible material as the knob is rotated in a tightening direction to tighten the article. In some embodiments, the body of the tightening mechanism may be integrally formed with one or more components of the base material. In a specific embodiment, the base material may comprise a shoe or a portion or component thereof, and the tightening mechanism and compressible material may be coupled with a heel portion of the shoe.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various embodiments are depicted in the accompanying drawings for illustrative purposes, and should in no way be interpreted as limiting the scope of the inventions.

(2) FIG. 1 is an isometric view of an example embodiment of a shoe that includes a reel-based tightening system

(3) FIG. 2 is a side view of the shoe of FIG. 1 with the concealing portion of the shoe in a first or uncompressed position.

(4) FIG. 3A is a side view of the shoe of FIG. 1 with the concealing portion of the shoe in a second or compressed position.

(5) FIG. 3B shows another example implementation of a shoe with a concealing portion having compressible portions on the sides of a tightening mechanism.

(6) FIG. 3C is another view of the shoe of FIG. 3B.

(7) FIG. 4 is a schematic cross-sectional view of an example embodiment of a tightening mechanism incorporated into an article and at least partially surrounded by a concealing portion.

(8) FIG. 5 is a schematic cross-sectional view of another example embodiment of a tightening mechanism incorporated into an article and at least partially surrounded by a concealing portion.

(9) FIG. 6A is a schematic cross-sectional view of another example embodiment of a tightening mechanism incorporated into an article and at least partially surrounded by a concealing portion.

(10) FIG. 6B is a schematic partially cross-sectional view showing an example embodiment of a

concealing portion having recesses or cutouts formed to allow a user to operate a tightening mechanism.

(11) FIG. 7 is a back view of an example embodiment of a boot having a tightening mechanism incorporated into the heel portion thereof.

(12) FIG. 8 is a side view of the boot of FIG. 7.

(13) FIG. 9 shows a side view of an example embodiment of a shoe with a concealing portion in an uncompressed position.

(14) FIG. 10A shows the shoe of FIG. 9 with the concealing portion in a compressed position

(15) FIG. 10B shows another example implementation of a shoe with a concealing portion.

(16) FIG. 11 is an exploded isometric view of a tightening mechanism.

(17) FIG. 12 shows a securing member and being coupled to an upper material of a shoe.

(18) FIG. 13 shows the securing member stitched to the upper material.

(19) FIG. 14 shows a housing being coupled to the securing member.

(20) FIG. 15 shows the housing and the securing member in an engaged configuration.

(21) FIG. 16 is a detailed view of the engagement members of the securing member and the housing.

(22) FIGS. 17-20 shows lace channels being applied to the upper material of the shoe.

(23) FIG. 21 shows a foxing layer of the shoe with a foam spacer applied thereto.

(24) FIG. 22 shows the foxing layer being applied to the shoe.

(25) FIG. 23A shows a cross-sectional view of the foam spacer.

(26) FIG. 23B shows a cross-sectional view of another example embodiment of a foam piece that can be used with some embodiment discussed herein.

(27) FIG. 23C shows a cross-sectional view of another example embodiment of a foam piece that can be used with some embodiment discussed herein.

(28) FIG. 24 shows a schematic cross-sectional view of an example embodiment of a tightening mechanism at least partially surrounded by a concealing portion in an uncompressed state.

(29) FIG. 25 shows a schematic cross-sectional view of a tightening mechanism with a concealing portion in a compressed state.

(30) FIG. 26 shows a schematic cross-sectional view of a tightening mechanism with a concealing portion having areas with different levels of compressibility.

(31) FIG. 27 shows a schematic cross-sectional view of a tightening mechanism with a concealing portion having a recess formed therein.

(32) FIG. 28 shows a schematic cross-sectional view of a tightening mechanism with a concealing portion having grooves formed therein.

(33) FIG. 29 shows a schematic cross-sectional view of a tightening mechanism with a concealing portion having cavities formed therein.

(34) FIG. 30 shows a schematic cross-sectional view of a tightening mechanism with a shielding element.

(35) FIG. 31 shows a schematic cross-sectional view of a tightening mechanism with a concealing portion that encloses a compressible material.

(36) FIG. 32A shows a schematic cross-sectional view of a tightening mechanism with a concealing portion that includes an exposed compressible material.

(37) FIG. 32B shows an example implementation of a tightening mechanism and concealing portion.

(38) FIG. 32C shows yet another example implementation of a tightening mechanism 508 and concealing portion.

(39) FIG. 33 is an exploded isometric view of a tightening mechanism.

(40) FIG. 34 shows a securing member being coupled to an upper material of a shoe.

(41) FIG. 35A shows a foxing layer and spacer being applied over the securing member.

(42) FIG. 35B shows a lace channel being applied to the upper material.

(43) FIG. 35C shows the assembly being back-part molded.

(44) FIG. 36 shows a housing being coupled to the securing member.

(45) FIG. 37A shows a spool and knob being coupled to the housing.

(46) FIG. 37B shows an example embodiment having a single piece that incorporates a securing member and a housing.

(47) FIG. 37C shows a foaming layer being applied over the single piece that incorporates the securing member and the housing.

(48) FIG. 38 is a schematic cross-sectional view of a tightening mechanism and

(49) concealing portion taken in a plane that intersects shielding elements.

(50) FIG. 39 is a schematic cross-sectional view of the tightening mechanism and concealing portion taken in a plane in which the concealing portion has a reduced height that is lower than in the plane of FIG. 38.

(51) FIG. 40 is a schematic cross-sectional view of the tightening mechanism and concealing portion in which the concealing portion can be compressed.

(52) FIG. 41 is an exploded view of an example implementation of a tightening mechanism and a concealing portion.

(53) FIG. 42 shows the assembled tightening mechanism and concealing portion of FIG. 41.

(54) FIG. 43 is a side view of the tightening mechanism and concealing portion of FIG. 41.

(55) FIG. 44 is a side view of a shoe having a tightening mechanism and a concealing portion at least partially surrounding the tightening mechanism.

(56) FIG. 45 shows a shaping member with a housing of the tightening mechanism mounted thereto.

(57) FIG. 46 is a cross-sectional view of the shoe of FIG. 44 showing the concealing portion and the housing coupled to the shoe.

(58) FIG. 47 is a side view of a shoe having a tightening mechanism and a concealing portion at least partially surrounding the tightening mechanism.

(59) FIG. 48 shows another view of the shoe of FIG. 47.

(60) FIG. 49 shows a spacer that can be configured to provide the shape of the concealing portion of FIG. 47.

(61) FIG. 50 is an isometric view of a boot having a tightening mechanism mounted onto the tongue of the boot and a concealing portion at least partially surrounding the tightening mechanism.

(62) FIG. 51 is a side view of the boot of FIG. 50.

(63) FIG. 52 is a detailed view of the concealing portion and tightening mechanism on the boot of FIG. 50.

(64) FIG. 53 shows a user actuating the tightening mechanism of the boot of FIG. 50.

(65) FIG. 54 shows a wrist brace having a tightening mechanism and a concealing portion at least partially surrounding the tightening mechanism.

(66) FIGS. 55a-c show a housing of a tightening mechanism being coupled with a foam backing material, which is in turn coupled with a shoe or other apparel.

(67) FIGS. 56a-b show a housing of a tightening mechanism being an integral component of a heel counter of a shoe.

(68) FIGS. 57a-d show a cover plate that is positionable over a housing and knob of a tightening mechanism.

(69) FIG. 58 shows a housing of a tightening mechanism integrally formed with an outsole of a shoe.

(70) FIGS. 59a-b show a housing of a tightening mechanism integrally formed with an outer material that is coupled with a shoe.

(71) FIGS. 60a-c show a flexible strip of material coupled with a shoe so as to be positioned over a tightening mechanism to hide a portion of the tightening mechanism from view of a user.

- (72) FIG. **61** is a perspective view of an embodiment of a lacing system in use with a sport shoe.
- (73) FIG. **62** is a perspective view of an embodiment of a lacing system.
- (74) FIG. **63** is an exploded perspective view of the reel from the lacing system of FIG. **62**.
- (75) FIG. **64** is another exploded perspective view of the reel of FIG. **63**.
- (76) FIG. **65** is a side view of the reel of FIG. **63** with the knob member shown in a disengaged position drawn in normal lines, and with the knob member in an engaged position shown drawn in dotted lines.
- (77) FIG. **66** is a perspective view of the base member from the reel of FIG. **63**.
- (78) FIG. **67** is a top view of the base member of FIG. **64**.
- (79) FIG. **68** is a bottom view of the base member of FIG. **64**.
- (80) FIG. **69** is a cross sectional side view of the base member of FIG. **64**.
- (81) FIG. **70A** is perspective view of the spool member from the reel of FIG. **63**.
- (82) FIG. **70B** is a perspective view of another embodiment of a spool member.
- (83) FIG. **71** is another perspective view of the spool member of FIG. **70A**.
- (84) FIG. **72** is a side view of the spool member of FIG. **70A**.
- (85) FIG. **73A** is a cross sectional view of the spool member of FIG. **70A** shown with a lace secured thereto in a first configuration.
- (86) FIG. **73B** is a cross sectional view of the spool member of FIG. **70A** shown with a lace secured thereto in a second configuration.
- (87) FIG. **73C** is a perspective view of the spool member of FIG. **70A** showing a lace being secured to the spool member in a third configuration.
- (88) FIG. **73D** is a perspective view of the spool member of FIG. **70A** showing the lace being secured to the spool member in a fourth configuration.
- (89) FIG. **74** is a top view of the spool member of FIG. **70A** shown disposed in the housing of the base member of FIG. **64**.
- (90) FIG. **75** is an exploded perspective view of the knob member from the reel of FIG. **63**.
- (91) FIG. **76** is another exploded perspective view of the knob member from FIG. **75**.
- (92) FIG. **77** is a perspective view of a pawl from the knob member of FIG. **75**.
- (93) FIG. **78** is another perspective view of the pawl from the FIG. **77**.
- (94) FIG. **79** is a top view of the pawls of FIG. **75** disposed in the knob core of FIG. **75**, with the pawls configured to engage the housing teeth of the housing.
- (95) FIG. **80** is a top view of the pawls of FIG. **75** shown engaged with the housing teeth on the base member of FIG. **64**.
- (96) FIG. **81** is a top view of the pawls of FIG. **75** shown displaced radially inwardly as the knob member is rotated in the tightening direction.
- (97) FIG. **82** is a top view of the spring bushing, fastener, and knob spring of FIG. **75** shown assembled with the knob core of FIG. **75**.
- (98) FIG. **83A** is an exploded view of the reel of FIG. **64** shown in an engaged configuration.
- (99) FIG. **83B** is a cross sectional view of the reel of FIG. **64** shown in an engaged configuration.
- (100) FIG. **84A** is an exploded view of the reel of FIG. **64** shown in a disengaged configuration.
- (101) FIG. **84B** is a cross sectional view of the reel of FIG. **64** shown in a disengaged configuration.
- (102) FIG. **85** is a perspective view of an alternative embodiment of a base member that can be used in place of the base member of FIG. **64**.
- (103) FIG. **86** is a cross sectional view of an alternative embodiment of a knob core.

DETAILED DESCRIPTION OF THE INVENTION

(104) FIG. **1** is an isometric view of an example embodiment of a shoe **100** that includes a reel-based tightening system. Although many embodiments are discussed herein as relating to shoes or other footwear, the embodiments disclosed herein can also related to other types of wearable articles, and to other objects that can be tightened and/or loosened (e.g., boots, hats, belts, sandals,

gloves, braces, backpacks, snowboard bindings). The shoe **100** of FIG. **1** can include a first portion **102a** and a second portion **102b** that can be drawn towards each other to tighten the shoe **100** and can be moved away from each other to loosen the shoe **100**. The first and second portions **102a** and **102b** can be spaced apart forming a gap **104** therebetween, or, in some embodiments, the first and second portions **102a** and **102b** can touch or overlap. A tension member, such as a lace **106**, can extend between the first and second portions **102a** and **102b** so that increased tension on the lace **106** can cause the first and second portions **102a** and **102b** to be drawn together, and so that reducing tension on the lace **106** can cause the first and second portions **102a** and **102b** to move apart from each other. The lace **106** can be coupled to a tightening mechanism **108** that is configured to adjust the tension on the lace **106** for tightening and/or loosening the shoe **100**. The shoe **100** can include one or more lace guides **110** configured to direct the lace **106** along a lace path between the first and second portions **102a** and **102b** of the shoe **100**. Although many embodiments are disclosed as using a lace **106**, other tensioning members (e.g., a strap) can be used for the various embodiments disclosed herein.

(105) The tightening mechanism **108** can be mounted onto the heel portion of the shoe **100**, as shown in FIG. **1**, or to various other portions of the shoe **100**, such as, for example, to the tongue or to a side portion of the shoe **100**. The shoe can also include one or more lace channels **112** configured to direct the lace **106** to the tightening mechanism **108**, and the lace channels **112** can be positioned at least partially under an outer layer of the shoe **100** so that the lace channels **112** are at least partially hidden from view.

(106) The tightening mechanism **108** can be at least partially concealed or protected by a concealing portion **114** of the shoe **100** that at least partially surrounds the tightening mechanism **108**. In some embodiments, the concealing portion **114** can include a compressible area that allows the concealing portion **114** to be transitioned between a first, uncompressed position, as shown in FIG. **2**, to a second, compressed position, as shown in FIG. **3A**. The concealing portion **114** can be compressible around substantially the full circumference of the tightening mechanism **108**, or at only certain portions around the tightening mechanism **108**. For example, in some embodiments, the concealing portion **114** can be compressible on right and left sides (e.g., at 3- and 9-o'clock) of the tightening mechanism **108** and can be substantially incompressible at the areas below and/or above (e.g., at 6- and 12-o'clock) the tightening mechanism **108** (e.g., as shown in FIGS. **3B** and **3C**). In other embodiments, the concealing portion **114** can be compressible at the areas below and/or above (e.g., at 6- and 12-o'clock) the tightening mechanism **108** and can be substantially incompressible on right and left sides (e.g., at 3- and 9-o'clock) of the tightening mechanism **108**. In the uncompressed position shown in FIG. **2**, the concealing portion **114** of the shoe **100** can surround at least a portion of the tightening mechanism **108** to at least partially hide the tightening mechanism **108** from view, which can improve the visual appearance of the shoe **100**. For example, for certain types of wearable articles (e.g., some golf shoes, running shoes, and casual shoes), the presence of an exposed tightening mechanism **108** can appear bulky or otherwise be inconsistent with the style of the article. Also, in some embodiments, the undesirable look of an exposed tightening mechanism **108** is further compounded on smaller sized shoes. By at least partially concealing the tightening mechanism **108**, the concealing portion **114** of the shoe **100** can increase the aesthetic appeal of the shoe **100**.

(107) Protecting or partially concealing the tightening mechanism **108** with a substantially resilient concealing portion **114** can allow aesthetically pleasing incorporation of the tightening mechanism **108** with the article. For example, as shoe sizes change, there can be a substantial dimensional reduction in the mounting area in the heel portion of the shoe (e.g., the shoe sizes get smaller). A substantially resilient concealing portion **114** can be formed around various surfaces to produce a visually appealing final structure that may not be possible with an entirely rigid shielding mechanism. As discussed elsewhere herein, the concealing portion **114** can incorporate some rigid components while still permitting adaptation to different sized areas.

(108) In some embodiments, the concealing portion **114** can protect the tightening mechanism **108** from damage and/or unintentional actuation. For example, an exposed tightening mechanism **108** can be unintentionally actuated when, for example, the tightening mechanism **108** is struck during contact sports. In some embodiments, unintentional actuation of the tightening mechanism **108** can unintentionally loosen the lace **106** or can over-tighten the lace **106**, which can cause discomfort and can degrade the performance of an athlete. By at least partially concealing the tightening mechanism **108**, the concealing portion **114** of the shoe **100** can protect the tightening mechanism **108** from being unintentionally actuated or damaged.

(109) When a compressing force (shown schematically by arrows in FIG. 3A) is applied to the concealing portion **114**, the collapsible area can collapse thereby transitioning the concealed portion **114** to the second or collapsed position, thereby increasing the amount of the tightening mechanism **108** that is exposed. The tightening mechanism **108** can include a rotatable knob **116** that is configured to be rotatable about an axis **118**. Rotation of the knob **116** in a tightening direction (e.g., clockwise) can tighten the shoe **100**, for example, by gathering lace **106** around a rotatable spool (not shown). In some embodiments, rotation of the knob **116** in a loosening direction (e.g., counterclockwise) can loosen the shoe **100**, for example, by releasing lace **106** from the spool. In some embodiments, the knob can be rotated between 60° and 180° degrees in the loosening direction to release the lace **106** from the spool. In some embodiments, the knob **116** can be configured to be pulled axially outwardly along the direction of the axis **118** to release tension on the lace **106**. In some embodiments, actuation of the knob **116** (e.g., rotation in the loosening direction or pulling in axial direction) can allow the spool to rotate freely independent of the knob **116**, which can allow for rapid loosening of the shoe **100**. In some of these rapid loosening embodiments, it can be especially advantageous to protect the knob **116** to prevent accidental actuation, which can cause accidental rapid loosening.

(110) In the compressed position, the concealing portion **114** of the shoe **100** can expose a sufficient portion of the knob **116** to allow a user to actuate the knob **116**, such as by rotating the knob **116** in a tightening direction, or in a loosening direction, or by pulling the knob **116** axially outwardly. The compressible area can be configured to compress (e.g., axially in the direction of the axis **118**) under pressure applied by the fingers of the user, and in some embodiments, the compressible area can have sufficient resistance to protect against unintentional actuation of the knob **116**. The compressible area can be resilient such that the concealing portion **114** returns to the first or uncompressed position when the compressing force is removed.

(111) The concealing portion **114** of the article (e.g., the shoe **100**) can radially surround at least a portion of the knob **116**. As shown schematically in FIG. 4, when the concealing portion **114** is in the uncompressed position, at least a portion of the knob **116** can be disposed axially rearward of an outer surface **120** of the concealing portion **114** in the direction of the axis **118**. As used herein the term “rearward” is used broadly to mean that one object, or portion thereof, is displaced back from another object, or portion thereof, even if the first object, or portion thereof, is not positioned directly behind the other object, or portion thereof. Also, in many instances, the terms “rearward,” “forward,” “inward,” “upward,” “top,” “bottom,” and the like can be used to describe locations or directions based on the orientation of the tightening mechanism, regardless of the orientation that the tightening mechanism has to article or the surrounding environment. Thus, at least a portion of the knob **116** can be disposed axially rearward of the outer surface **120** of the concealing portion **114** even when the concealing portion **114** does not cover the top surface **122** of the knob **116**. The top surface **122** of the knob **116** can be uncovered, for example, such that the top surface **122** of the knob **116** is visible when viewed from the top down. In some embodiments, a majority of the knob **116** can be disposed rearward of the outer surface **120** of the concealing portion **114**. In some embodiments, the entire, or substantially the entire, knob **116** can be disposed rearward of the outer surface **120** of the concealing portion **114**. For example, in some embodiments, the top surface **122** of the knob **116** can be substantially flush with the outer surface **120** of the concealing portion **114**,

as shown in FIG. 5. The concealing portion **114** can extend upward at least as far as the top of the sides **124** of the knob **116**, or at least past the lower surface of the knob **116**. In some embodiments, the sides **124** of the knob **116** can be partially, entirely, or substantially entirely, rearward of the outer surface **120** of the concealing portion **114**. In some cases, a portion of the top **122** of the knob **116** can extend forward of the outer surface **120** of the concealing portion **114** (e.g., due to a generally frusta-conical shape, a curved shape, or other contours, of the top **122** of the knob **116**), as shown in FIG. 6A. Various configurations are possible. For example, in some embodiments, at least about 95%, at least about 90%, at least about 85%, at least about 80%, or at least about 75% of the rotatable knob **116** (or of the entire tightening mechanism **108**) can be disposed rearward of the outer surface **120** of the concealing portion **114**.

(112) The concealing portion **114** can have a recess **126**, and the tightening mechanism **108** can be disposed in the recess **126**. In some embodiments, the recess **126** can extend only partially through the article. For example, a base layer **128** of the article can be located at the bottom of the recess **126**, and the tightening mechanism **108** can be secured to the base layer **128**. A housing **130** of the tightening mechanism **108** can be attached to the base layer **128**, for example, by stitching, rivets, adhesive, or other suitable manner. The concealing portion **114** can be attached to the base layer **128**. In some embodiments, the concealing portion **114** can be one or more additional layers applied to the outside of an otherwise completed article, while in other embodiments, the concealing portion **114** can be formed as an integral portion of the article. In some embodiments, the recess **126** can extend through the article (e.g., through the heel wall, or side wall, of the shoe **100**).

(113) FIG. 6B is a schematic partially cross-sectional view showing an example embodiment of a concealing portion **114** having recesses or cutouts formed to allow a user to operate a tightening mechanism **108**. The left side of FIG. 6B shows a side view of the tightening mechanism **108** and concealing portion **114**. The right side of FIG. 6B shows a cross-sectional view through a center of the tightening mechanism **108**, and the cross-sectional portion of FIG. 6B is shown having cross-hatching to emphasize the cross-sectional portion. As can be seen in FIG. 6B, and as discussed elsewhere herein, the concealing portion **114** can have areas **114a** and **114b** that extend higher than other areas **114c** of the concealing portion **114**. More of the tightening mechanism **108** can be exposed at the lower areas **114c** of the concealing portion **114**, for example, to allow a user to grip the sides of the tightening mechanism **108** (e.g., during tightening or loosening of the system). In some embodiments, a recesses, cutout, or scalloped area, etc. can form the lower portions **114c** of the concealing portion **114**. In some embodiments, the higher areas **114a** and **114b** of the concealing portion **114** can provide more protection and/or concealment than the lower areas **114c**. In some embodiments, the higher areas **114a** and **114b** can be positioned above and below the tightening mechanism **108** (e.g., at 6- and 12-o'clock), while the lower portions **114c** can be positioned on the sides of the tightening mechanism **108** (e.g., at 3- and 9-o'clock). In some embodiments, the concealing portion **114** can be compressible at the lower portions **114c**, and can be substantially incompressible at the higher portions **114a** and **114b**. In some embodiments, the concealing portion **114** (including the areas **114a**, **114b**, and **114c**) can be substantially incompressible, and the lower portions **114c** can allow the user to actuate the tightening mechanism **108** without displacement of the concealing portion **114**. For example a rigid material (e.g., a rigid foam or plastic) can surround at least part of the tightening mechanism **108** to form the shape of the concealing portion **114**.

(114) FIG. 7 is a back view of an example embodiment of a boot **200** having a tightening mechanism **208** incorporated into the heel portion thereof. FIG. 8 is a side view of the boot **200**. The boot **200** can have features similar to, or the same as, the shoe **100**, or the other embodiments described herein. The tightening mechanism **208** can be positioned at or near the collar of the boot **200**. The concealing portion **214** can completely surround the tightening mechanism **208** by a full 360 degrees, as shown in FIG. 7, or the concealing portion **214** can surround only a portion of the

tightening mechanism **208** (e.g., by at least about 90 degrees, at least about 180 degrees, at least about 270 degrees, at least about 300 degrees, or at least about 330 degrees). In some embodiments, the concealing portion **114** can surround the areas of the tightening mechanism **208** that are most susceptible to being struck during use (e.g., the below the tightening mechanism **208** between the tightening mechanism and the sole of the shoe).

(115) FIGS. **9** and **10** shows side views of an example embodiment of a shoe **300**, which can have features similar to the shoe **100**, the boot **200**, or the other embodiments disclosed herein. FIG. **9** shows a concealing portion **314** in an uncompressed position, and FIG. **10A** shows the concealing portion **314** in a compressed position. A tightening mechanism **308** can be mounted onto the heel portion of the shoe **300**. As can be seen in FIG. **9**, the concealing portion **314** can cover, or substantially cover, the sides of the knob **316** at a first area **314a** (e.g., below the tightening mechanism **308** or between the tightening mechanism **308** and the sole of the shoe **300**) and/or at a second area **314b** (e.g., above the tightening mechanism **308** or between the tightening mechanism **308** and the collar of the shoe **300**). The second area **314b** can be positioned generally on an opposite side of the tightening mechanism **308** from the first area **314a**. Thus, in some embodiments, a cross-sectional view of the shoe **300** taken through the axis **318** and in the plane of the page can be similar to FIGS. **5-6** with respect to the positioning of the knob **316** and the concealing portion **314**. Accordingly, the discussion of FIGS. **5-6** can be applied to the shoe **300**, in some embodiments.

(116) With further reference to FIG. **9**, the concealing portion **314** can cover only a portion of the sides of the knob **316** at a third area **314c** (e.g., on a left side of the tightening mechanism **308**) and/or at a fourth area **314d** (e.g., on a right side of the tightening mechanism **308** (hidden from view in FIG. **9**)). The fourth area **314d** can be positioned generally on an opposite side of the tightening mechanism **308** from the third area **314c**. Thus, in some embodiments, a cross-sectional view of the shoe **300** taken through the axis **318** and transverse to the plane of the page can be similar to FIG. **4** with respect to the position of the knob **316** and the concealing portion **314**. Accordingly, the discussion of FIG. **4** can be applied to the shoe **300**, in some embodiments. A portion of the knob **316** can be partially exposed, for example, on the right and left sides at the areas **314c** and **314d**. The partially exposed knob **316** can facilitate gripping of the knob **316** when the user actuates the knob **316**.

(117) With reference to FIG. **10A**, at least portions of the concealing portion **314** can be compressible to a compressed position to increase the amount of the knob **316** that is exposed, thereby facilitating the gripping of the knob **316** when the user actuates the knob **316**. In some embodiments, the areas **314c** and/or **314d** can be more compressible than the areas **314a** and/or **314b**. For example, in some embodiments, one or both of the areas **314a** and/or **314b** can be substantially incompressible, for example, having a rigid protective member disposed therein to protect the tightening mechanism **308** from being struck near the areas **314a** and/or **314b**. In some embodiments, the shoe **300** can be configured to have the open-side configuration shown in FIG. **10B** when at rest, without the concealing portion **314** being compressed. In some embodiments, the concealing portion **314** (including the areas **314a-d**) can be substantially incompressible. The at least partially open sides of the embodiment shown in FIG. **10B** can allow a user to manipulate the tightening mechanism **308** without displacing the concealing portion **314**.

(118) FIG. **11** is an exploded isometric view of a tightening mechanism **408**, which can be used with the shoe **100**, the boot **200**, the shoe **300**, or the other embodiments disclosed herein. The tightening mechanism **408** can include a housing **432**, a securing member **434**, a spool **436**, and a knob **416**. The spool **436** can be mounted into the housing **432** such that the spool **436** is rotatable about the axis **418**. The housing **432** can have one or more lace holes **438a** and **438b** configured to receive the lace into the housing **432**, so that the lace can be coupled to the spool **436** so that rotation of the spool **436** in a tightening direction gathers the lace into a channel **440** in the spool **432**. The spool **436** can include teeth **442** configured to engage teeth (hidden from view) on an

underside of the knob **416**, so that rotation of the knob **416** can cause rotation of the spool **436**, thereby allowing a user to tighten the lace by rotating the knob **416**. The housing can include teeth **444** that are configured to engage pawls (hidden from view) on the underside of the knob **416** such that the knob **416** is prevented from rotating in a loosening direction and permitted to rotate in a tightening direction. In some embodiments, the knob **416** can be lifted axially away from the housing **432** to a disengaged position that allows loosening of the lace. Many other configurations can be used for the tightening mechanism **408**.

(119) With reference now to FIGS. **12** and **13**, the securing member **434** can be secured to the article. For example, an upper material **446** of a shoe can have a hole **448** formed in the heel portion thereof. The securing member **434** can be inserted into the hole **448** from the inside of the upper material **446** back towards the heel portion thereof, as shown in FIG. **12**. The securing member **434** can have side walls **450** that surround an opening **452**. In some embodiments, the side walls **450** can extend through the hole **448**, and in some cases can stretch the upper material **446** to fit around the side walls **450**. The securing member **434** can have a securing flange **454**, which can remain on the inside of upper material **446** (shown in phantom lines in FIG. **13**). The securing flange **454** can be secured upper material **446**, such as by stitching **456**, or by rivets, or an adhesive, or any other suitable manner. The securing member can include a shield element **458** configured to extend out to cover a side portion of the knob **416**, when the tightening mechanism **408** is assembled. The shield element **458** can be positioned on a lower side of the tightening mechanism **408** so that the shield element **458** is positioned between the knob **416** and the sole of the shoe once assembled. Thus, the shield element **458** can provide protection against striking the knob **416** from below (e.g., such as may occur when walking down stairs or during contact sports).

(120) With reference now to FIGS. **14-16**, the housing **432** can be attached to the securing member **434**. For example, the securing member **434** can have one or more engaging members **460a** and **460b** that are configured to engage with one or more corresponding engaging members **462a** and **462b** on the housing **432**. The engaging members **460a** and **460b** can engage the engaging members **462a** and **462b** by a snap-fit connection, a friction-fit connection, a clasp, or any other suitable manner. For example, the engaging members **460a** and **460b** on the securing member **343** can include protrusions that fit into notches **462a** and **462b** in the housing **432** to snap the housing into the secured position. Other configurations are possible. In some embodiments, the housing **432** can be removably attached to the securing member **434** so that the housing **432** can be removed, for example, if the tightening mechanism **408** is to be repaired or replaced or cleaned.

(121) With reference to FIGS. **17-20**, which show the upper material **446** from a bottom view, lace channels **412a** and **412b** can be installed to direct the lace to the tightening mechanism **408**. The lace channels **412a** and **412b** can be positioned inside the upper material **446** so that they are hidden from view once the shoe is fully assembled. Lace ports **464a** and **464b** can be positioned to receive the lace, for example, at an end of the gap between the first and second portions of the shoe. The lace channel tubes **412a** and **412b** can be coupled to the lace ports **464a** and **464b** and to the lace holes **438a** and **438b**, for example, by inserting the tubes **412a** and **412b** into the lace ports **464a** and **464b** and into the lace holes **438a** and **438b**. Adhesive backing tape **466** can be placed over the tubes **412a** and **412b** to hold them in place. An adhesive can be applied over the lace channel tubes **412a** and **412b** (e.g., onto the backing tape **466**), and padding strips **468a** and **468b** can be adhered over the lace channel tubes **412a** and **412b** by the adhesive. The padding strips **468a** and **468b** can reduce discomfort caused by the tubes **412a** and **412b** pressing on the foot of a wearer when in use, and can also hide the shape of the tubes **412a** and **412b**. In some embodiments, the lace channels **412a** and **412b** can extend only partially across the collar of the shoe so that the lace can exit at locations on the side of the collar (e.g., at or near the midpoint of the collar). For example, FIGS. **1-3A** show an example embodiment in which the lace extend outside the shoe across a portion of the collar and then enters the lace channels that guide the lace under the shoe material to the tightening mechanism. This configuration can allow for collar compression,

simplified assembly, flexibility, and can eliminate pressure points, in some embodiments.

(122) With reference now to FIGS. 21 and 22, the concealing portion **416** of the shoe can be formed to conceal and/or to protect the tightening mechanism **408**. In some embodiments, a foxing or outer layer **470** can be cut to a shape that is suitable to fit the contours of the article (e.g., the heel portion of the shoe **400**). A compressible material, such as a foam **472** can be applied to the inside surface of the outer layer **470**, such as by applying an adhesive, such as a polyurethane thermoplastic adhesive (e.g., Bemis brand 3206D polyurethane thermoplastic adhesive (e.g., 6 mil (0.006 inches), although other thicknesses can be used depending on the materials used and the intended use of the article)). Other adhesives can be used depending on the materials used and the intended use of the article. The outer layer **470** can be applied to the shoe **400**, as shown in FIG. 22, for example. An adhesive can be applied (e.g., sprayed on) to the inside surface of the outer layer **470** and the outer layer **470** can be pressed against the underlying portions of the article (e.g., to the upper material **446**). In some embodiments, a single application of an adhesive to the inside surface of the outer layer **470** can be used for adhering the foam **472** to the layer **470** and to adhere the layer **470** to the shoe **400**. In other embodiments, separate adhesives and/or separate applications of the adhesive can be used for attaching the foam **472** and for attaching the layer **470** to the shoe **400**. In some embodiments, the foam **472** can be attached (e.g., adhered) to the shoe **400** directly (e.g., over the upper material **446**), and in some cases the layer **470** can then be applied over the top of the foam **472**.

(123) A hole **426** can extend through the layer **470** and the foam **472** and can be configured to receive the tightening mechanism **408** therein when the layer **470** is applied to the shoe **400**. If a spray adhesive is applied to the inside surface of the layer **470**, the hole can be masked off during application of the adhesive. Also, the foam **472** and/or the layer **470** surrounding the hole **426** can be colored (e.g., painted or dyed) so that it resembles the color and/or style of the outer appearance of the shoe **400**. The foam **472** and/or the layer **470** can come in the color that matches or resembles the color of the shoe **400**, or can be color matched, e.g., using dye additives. Also, the knob **416** or other components of the tightening mechanism **408** can have a color that is the same as, or similar to, the color and/or style of the outward appearance of the shoe **400** (e.g., to deemphasize the visual appearance of the tightening mechanism **408**). The layer **470** can also be stitched to the shoe **400**, or attached to the shoe **400** by other suitable manners.

(124) The outer layer **470** and the foam **472** can have different shapes for different sizes and styles of shoes and for different types of articles. The foam **472** can have a shape and thickness configured to raise the outer layer **470** away from the underlying layer **446** by a height that is sufficient to cover part of, a majority of, substantially all of, or all of the sides of the knob **416**, as discussed herein. In some embodiments, the layer **470** can be made from a polyurethane-backed nylon fabric, such as polyurethane-backed Cordura® fabric, which can have a low friction nylon interface that allows the user's fingers to slide easily across the surface of the layer **470** when turning the knob **416**. Other low friction materials can also be used. In some embodiments, materials can be modified to add a low friction interface around the perimeter of the tightening mechanism. For example, direct injection molding, radio frequency welding, or debossing can be used to create the low friction interface. In some embodiments, a cover piece can be disposed around at least a portion of the tightening mechanism and can secure the fabric of the cover layer **470** (e.g., to the tightening mechanism). For example, a ring made of plastic (or other suitable material) can surround at least a portion of the tightening mechanism, and, in some embodiments, can form a low friction interface to allow a user's fingers to slide smoothly when operating the tightening mechanism.

(125) FIG. 23A is a cross-sectional view of an example embodiment of a compressible member or foam piece **472** that can be used with some embodiments. The sides **425a** and **425b** of the foam piece **472** can be configured to wrap around the heel of the shoe and onto the side portions of the shoe. The sides **425a** and **425b** can be tapered to form a smooth transition at the ends of the foam

piece 472 when mounted onto the shoe. The foam piece 472 can include the hole 426 therein. In some embodiments, the inside of the ring can chamfer outward to account for the curvature of the heel of the shoe 400. The foam piece 472 can be made from a variety of materials, such as, for example, Rubberlite V0525 Viso-Cel® slow rebound foam. Other open celled polyurethane foams can also be used, as well as other compressible materials. FIG. 23B shows a cross-sectional view of another example embodiment of a foam piece that can be used with some embodiment discussed herein. FIG. 23C shows a cross-sectional view of another example embodiment of a foam piece that can be used with some embodiment discussed herein. Various shapes of spacers (e.g., foam pieces 472) can be used depending on the shape and size of the article. For example, the embodiments of FIGS. 23B and 23C can have shorter side portions 425a and 45b than the embodiment of FIG. 23A, and the embodiment of FIG. 23C can have thinner ends on the side portions 425a and 425b than the embodiment of FIG. 23B.

(126) FIG. 24 is a cross-sectional view of an example embodiment of a tightening mechanism 518 incorporated into an article, such as the shoe 100, the boot 200, the shoe 300, the shoe 400, or the other embodiments disclosed herein. The tightening mechanism 508 can include a housing 532, a spool 536, and a knob 516, similar to the tightening mechanism 408 described herein. The housing 532 can be mounted to a base material 546, such as the heel counter or upper material of a shoe. In some embodiments, the housing 532 can be attached directly to the base material 546 (as shown in FIG. 24), such as by stitching through a securing flange 554 of the housing 532, or by rivets, or by an adhesive, or other suitable manner. In some embodiments, the housing 532 can be coupled to the article using a securing member (e.g., similar to the securing member 434 discussed herein). In some embodiments, the base material 546 can include a hole therein for receiving the housing 532, such that a portion of the housing 532 is disposed rearward of the base material 546, thereby reducing the height by which the tightening mechanism 508 extends forward of the base material 546, which can facilitate the concealment of the tightening mechanism 508, and can reduce the height of the concealing area 514, which can improve the visual appearance of the article.

(127) In some embodiments, padding 574 can be positioned rearward of the tightening mechanism 508 to provide comfort to the wearer and to prevent the tightening mechanism 508 from pressing against the portion of the wearer's body that contacts the article. For example, the tightening mechanism 508 can be incorporated into the tongue of a shoe or into a padded strap of a backpack or into other padded portions of wearable articles. In some embodiments, liners and other layers can be disposed rearward of the tightening mechanism 508, but are not shown in FIG. 24 for simplicity.

(128) A concealing portion 514 can at least partially surround the tightening mechanism 508. The concealing portion 514 can include a compressible area 576, which can be a foam material, as discussed herein. FIG. 24 shows the concealing portion 514 in an uncompressed position, and FIG. 25 shows the concealing portion 514 in a compressed position in which the compressible area is compressed (e.g., by a compressing force applied by a user's fingers) to expose the knob 516. In some embodiments, the compressible area 576 can be disposed between the base material 546 and an outer layer 570. In some embodiments, some or all of the area surrounding the tightening mechanism 508 can be substantially incompressible. For example, the area 576 of FIG. 24 can include a substantially incompressible material (e.g., a rigid plastic material or a rigid foam material).

(129) In FIG. 26, the concealing portion 514 can include a first area 576a that is more compressible than a second area 576b. The more compressible area 576a can be positioned radially inward from the less compressible area 576b. For example, the more compressible area 576a can surround at least a portion of the tightening mechanism 508, and the less compressible area 576b can surround at least a portion of the more compressible area 576a. In some embodiments, both the first compressible area 576a and the second compressible area 576b can include compressible foam, and the foam of the first compressible area 576a can be of a lower density and higher compressibility

than the foam of the second compressible area **576b**. In some embodiments, the second area **576b** is substantially not compressible. The first compressible area **576a** can have a radial width of at least about 5 mm, at least about 10 mm, at least about 15 mm, no more than about 20 mm, between about 5 mm and 15 mm, and/or about 10 mm. In some embodiments, the first compressible area **576a** can be wide enough to allow a user's fingers to compress the first compressible area **576a** without directly applying a compressing force onto the second area **576b**. In some embodiments, the first compressible area **576a** can have a width that is small enough that a compressing force applied by a user's finger directly applies a compressing force to both the first area **576a** and the second area **576b**.

(130) In some embodiments, the compressible area **576** can include a recess **578a** configured to facilitate compression of the compressible area **676**. In some embodiments, the recess **578a** can be disposed directly behind a layer of the compressible material (e.g., foam), so that when a compressing force is applied, the layer of the compressible material can collapse down into the recess **578a** to expose the tightening mechanism **508**. In some embodiments, the recess **578b** can be tapered (e.g., as shown in the lower portion of FIG. 27) so that a portion of the collapsible area **576** nearer to the tightening mechanism **508** can collapse more easily and/or further than a portion of the collapsible area **576** that is radially further from the tightening mechanism **508**. In some embodiments, the recess can include one or more cutouts or grooves **578c** formed in the compressible material (as shown in FIG. 28). Multiple grooves **578c** can be included such that one or more extensions of the compressible material can extend between the grooves **578c**. In some embodiments, the grooves **578d** can be tapered (e.g., as shown in the lower portion of FIG. 28) so that a portion of the collapsible area **576** nearer to the tightening mechanism **508** can collapse more easily and/or further than a portion of the collapsible area **576** that is radially further from the tightening mechanism **508**. In some embodiments, the recess can include a cavity **578e** that is a volume surrounded on all sides by the compressible material (e.g., foam). In some embodiments, the recess can include multiple cavities **578e** and **578f** (as shown in the upper portion of FIG. 29). In some embodiments, the size or distribution of the plurality of cavities **578e** and **578f** can vary such that a portion of the collapsible area **576** nearer to the tightening mechanism **508** can collapse more easily and/or further than a portion of the collapsible area **576** that is radially further from the tightening mechanism **508**. Although the upper portion of FIG. 29 shows only two cavities **578e** and **578f** for simplicity of illustration, some embodiments can include a larger number of cavities formed in the compressible material. In some embodiments one or more individual cavities **578g** can be tapered (as shown in the lower portion of FIG. 29), so that a portion of the collapsible area **576** nearer to the tightening mechanism **508** can collapse more easily and/or further than a portion of the collapsible area **576** that is radially further from the tightening mechanism **508**.

(131) The various recess types **578a-578g** shown in FIGS. 27-29 can be used individually or can be combined with others of the recess types **578a-578g** to provide various alternative configurations. In some embodiments, a recess structures **578a-578g** can extend rotationally to form arcuate recesses that at least partially surround the tightening mechanism **508**.

(132) In some embodiments, the tightening mechanism **508** can include one or more shield elements **558**. The shield element **558** can be, for example, integrally formed with the housing **532**, or the shield element **558** can be a separate component from the housing **532**. The shield element **558** can be a rigid extension that covers at least part of the side of the knob **516**. The shield element **558** can be configured to protect to the knob **516**, as discussed elsewhere herein. Various embodiments disclosed herein (e.g., the embodiments of FIGS. 24-29 and 31-32) can be modified to include a shield element **558** similar to that described in connection with FIG. 30. In some embodiments an additional shield element can be positioned generally opposite the shield element **558** shown in FIG. 30. For example, shield elements **558** can be positioned at about 6-o'clock and at about 12-o'clock, to provide protection to the tightening mechanism **508**, as discussed herein.

(133) In some embodiments, the compressible material **576** can be enclosed. For example, as

shown in the upper portion of FIG. 31, the base material 546 can wrap around the compressible material 576 such that the compressible material 567 is sandwiched between portions of the base material 546. In some embodiments, an outer layer 570 can extend around the compressible material 576 and can be coupled to the base material 546, as shown in the lower portion of FIG. 31, or the base material 546 can extend around the compressible material and can be coupled to the outer layer 570. The base material 546 and outer layer 570 can be coupled together, for example, by stitching, or rivets, or an adhesive, or any other suitable manner. In some embodiments, a layer separate from the base material 546 and the outer layer 570 can extend between the outer layer 570 and the base material 546 between the compressible material 576 and the knob 516, and the layer can be flexible so that it can be collapsed or displaced to expose the knob 516 (e.g., when a user applied a compressing force). The flexible layer can be positioned between the compressible material 576 and the knob 516, thereby separating the knob 516 from the compressible material 576, which can prevent the compressible material 576 from contacting the knob 516 when the compressible material 576 is deflected in the compressed state. If the deflected compressible material 576 contacts the rotatable knob it can interfere with rotation of the knob 516 and in some cases can become pinched by the knob 516. Thus, the layer separating the compressible material 576 from the knob 516 can prevent the compressible material 576 from interfering with operation of the knob 516.

(134) In some embodiments, the compressible material 576 can be uncovered, as shown in FIG. 32A. In some embodiments, slow recovery memory foam can be used as the flexible material 576, although various other compressible materials can also be used. In some embodiments, the top of the compressible material 576 can define the outer surface 520 of the concealing portion 514. The outer surface 520 of the compressible material 576 can be colored or patterned to coordinate with the color and/or styling of the article, thereby visually deemphasizing the concealing area 514.

(135) Many variations can be made to the embodiments disclosed herein. For example, in some embodiments, substantially incompressible guarding members (e.g., rigid plastic strips) can be insert molded into a compressible material to add rigidity and additional guarding to certain areas of the concealing portion 514 (e.g., the area below and/or above the tightening mechanism). For example, with reference to FIG. 26, in some embodiments, the first area 576a surrounding the tightening mechanism 508 can be substantially incompressible. For example, the first area 576a can include a guarding member (e.g., made of a rigid plastic material), which can be, for example, insert molded into the foam to create guards that protect and/or conceal the tightening mechanism 508.

(136) FIG. 32B shows an example implementation of a tightening mechanism 508 and concealing portion 514, which can have features similar to, or the same as, the embodiments shown in FIGS. 24-32A. In FIG. 32B, the tightening mechanism 508 can include a securing flange 554 that is flatter than those shown in FIGS. 24-32A. The size and shape of the securing flange 554, as well as the other features of the tightening mechanism 508 can vary depending on the size and shape of the article with which the tightening mechanism 508 is applied. For example, in FIG. 32B, the base material 546 can be, for example, a heel counter of a shoe, and the base material 546 can have a hole that receives a portion of the tightening mechanism 508 (e.g., a bottom of the housing 532) therein. In some embodiments, the base material 546 (e.g., heel counter) can be substantially flush with the bottom surface of the housing 532, as shown in FIG. 32B. Although not shown in FIG. 32B, padding or lining layers can be positioned rearward of the tightening mechanism 508, for example, to separate the tightening mechanism 508 from the wearer. The embodiment shown in FIG. 32B can be modified to incorporate the features shown and discussed in connection with FIGS. 24-32A.

(137) FIG. 32C shows another example implementation of a tightening mechanism 508 and concealing portion 514, which can have features similar to, or the same as, the embodiments shown in FIG. 24-32B. A housing 532 can be mounted onto a base material 546 (e.g., heel counter). In

some embodiments, the base material **546** (e.g., heel counter) does not include a hole that receive a portion of the housing **532** therein. The housing **532** can be secured (e.g., stitched or adhered) to the outside of the base material **546**. An outer material **570** can be elevated at the concealing portion **514**, e.g., by a spacer **576**, which can be a foam or plastic material, and can be compressible or substantially incompressible, as discussed herein. In some embodiments, additional foam can be used around the spacer **576**, such as collar foam **577** that surrounds a collar portion of a shoe. In some embodiments, a grommet **579** can surround all or a portion of the tightening mechanism **508**. The grommet **579** can be a ring. The grommet **579** can be positioned between the spacer **576** and the outer material **570**. In some embodiments, the outer material **570** can be stitched, adhered, or otherwise secured or coupled to the grommet **579**. The grommet **579** can be rigid or generally rigid, so that when the user presses down on the grommet **579**, it compresses an area of the concealing portion **514** positioned under the grommet **579**, which in some cases can be a full 360° area surrounding the tightening mechanism **508**, or a portion thereof.

(138) FIG. **33** is an exploded isometric view of a tightening mechanism **608**, which can be used with an article (e.g., the shoe **100**, the boot **200**, the shoe **300**, or other embodiments disclosed herein). The tightening mechanism **608** can include a housing **632**, a securing member **634**, a spool **636**, and a knob **616**. The spool **636** can be mounted into the housing **632** such that the spool **636** is rotatable with respect to the housing **632**. A lace can be coupled to the spool **636** so that rotation of the spool **636** in a tightening direction gathers the lace onto the spool **636**. The spool **636** can engage the knob **616**, so that rotation of the knob **616** can cause rotation of the spool **636**, thereby allowing the lace to be tightened by rotating the knob **616**. The knob **616** can include a top surface **622** and sides **624**. In some embodiments, the spool **636** and the knob **616** can be configured similarly to the spool **436** and knob **416** discussed above. Many other configurations can be used for the tightening mechanism **608**.

(139) The securing member can have side walls **650** that surround a recess **652**. The side walls **650** can have a first indented portion **651a** and a second indented portion **651b**, which can be positioned on generally opposite sides of the securing member **634** (e.g., on the right and left sides thereof). One or more holes or notches **641a** and **641b** can allow a lace to pass from outside the securing member **634** into the recess **652**. For example, notches **641a** and **641b** can be formed in the indented portions **651a** and **651b** of the side walls **650**. The securing member **634** can include engagement features (e.g., slots **643**) which can be configured to engage with engagement features (e.g., teeth **645**) on the housing **632** to allow the housing **632** to be secured to the securing member **634** (e.g., by a snap-fit engagement). The securing member **634** can include a securing flange **654**, which can extend radially outwardly from the base of the side walls **650**. In some embodiments, lace holes **638a** and **638b** are formed on the securing member **634** (e.g., on the bottom thereof), and lace channels can lead from the lace holes **638a** and **638b** to the notches **641a** and **641b** or holes that allow the lace to enter the recess **652**.

(140) The housing **632** can include side walls **655** and indented portions **657a** and **657b** which can align generally with the indented portions **651a** and **651b** of the securing member **634**. In some embodiments, internal side walls **647** surround a recess **659**. A gap can be formed between the side walls **655** and the internal side walls **647**. One or more notches **649a** and **649b** or holes can be formed in the side walls **655** (e.g., at the base of the indented portions **657a** and **657b**), and one or more notches **661a** and **661b** or holes can be formed in the internal side walls **647**. The notches or holes can allow the lace to pass into the recess **659**, and for example, can align with the holes or notches **641a** and **641b** formed in the securing member **634**.

(141) With reference to FIGS. **34** and **35A**, a securing member **634** can be secured to the article (e.g., to an upper material **646** of a shoe). For example, securing flange **654** can be stitched to the upper material **646**, or secured thereto by other suitable securing mechanisms. The upper material **646** can include one or more lace holes **633a** and **633b** which can align with the lace holes **638a** and **638b** on the securing member **634**. As shown in FIG. **35B**, lace channels **612**, similar to those

discussed in connection with FIGS. 17-20, can be applied inside the upper material **646** and can direct the lace to the lace holes **633a** and **633b** and to the securing member **634**. In some embodiments, the tightening mechanism **608** is disposed outside the upper material **646**, and the upper material **646** does not include a hole that allows a portion of the tightening mechanism to be disposed rearward of the upper material **646**.

(142) A foxing or outer layer **670** can be positioned over the securing member **634**. A spacer **676** can attach to the underside of the layer **670** (e.g., using an adhesive). The spacer **676** can be a compressible material, a rigid material, or a semi-rigid material. The spacer **676** can have a first or upper portion **676a** and a second or lower portion **676b** separated by gaps **653a** and **653b** or thinner portions of the spacer **676**. A hole can extend through the outer layer **670** and through the spacer **676**. The spacer **676** can be configured to fit around the outside of the side walls **650** of the securing member **634** when the layer **670** is mounted onto the article, and the gaps **653a** and **653b** in the spacer **676** can align with the indented portions **651a** and **651b** of the side walls **650** on the securing member **634**. In some embodiments, the gaps **653a** and **653b** can provide paths for the lace to pass through. In some embodiments, the spacer **676** can extend a full 360 degrees around the opening **626**, and the gaps **653a** and **653b** can be omitted. The hole **626** through the layer **670** and spacer **676** can align over the recess **652** when the layer **670** is mounted onto the article. In some embodiments, the assembly can be back part molded, as shown, for example, in FIG. 35C.

(143) As can be seen in FIG. 36, the housing **632** can be mounted onto the securing member **634**. In some embodiments, a portion **671** of the foxing or outer layer **670** surrounding the hole **626** can extend over the securing member **634** so that the portion **671** of the layer **670** is pressed down into the recess **652** of the securing member **634** when the housing **632** is inserted therein. In some embodiments, because the portion **671** of the layer **670** is be pinched between the housing **632** and the securing member **634**, there is no gap between the edges of the foxing layer **670** and the tightening mechanism **608**, which can prevent debris from entering a space around the tightening mechanism **608**.

(144) As discussed above, the housing **632** and the securing member **634** can include corresponding engagement features that are configured to secure the housing **632** to the securing member **634**, such as, for example, by a snap fit, a friction fit, etc. In some embodiments, the housing **632** can be removably attachable to the securing member **634**, so that the housing **632** can be removed (e.g., for repair, replacement, or cleaning). Because the housing **632** is inserted over the foxing layer **670**, the housing **632** can be removed from the securing member **634** without removing or cutting the foxing layer **670**.

(145) As shown in FIG. 37A, the spool **636** can receive a lace **606** and can be rotatably supported in the recess **659** of the housing **632**. The knob **616** can be rotatably mounted onto the housing **632** and can be configured such that rotating the knob **616** can tighten the lace **606** by causing the spool **636** to rotate. In some embodiments, the side walls **655** and/or the side walls **650** can surround at least a portion of the side **624** of the knob **616**, thereby forming rigid shielding elements that can protect the knob **616** from accidental actuation. The indented portions **657a** and **657b** and/or **651a** and **651b** can expose portions of the side **624** of the knob **616**, to allow a user to grip the sides **624** of the knob **616** (e.g., for tightening). A concealing portion **614** of the article can at least partially surround the sides **624** of the knob **616** to conceal or protect the tightening mechanism **608**. For example, the spacer **676** can press the foxing layer **670** up around the tightening mechanism **608**. In some embodiments, the concealing portion **614** can be higher at some areas surrounding the tightening mechanism **608** than at other surrounding areas.

(146) Many variations are possible. For example, with reference to FIG. 37B, in some embodiments, the housing **632** can be incorporated into the securing member **634**, for example, as a single integrally formed piece **632'** that can be attached directly to the article. The housing piece **632'** can combine the features of the housing **632** and the securing member **634** discussed above. Because the housing piece **632'** can be a single integral piece, the engagement features of the

securing member **634** and housing **632** can be omitted in the housing piece **632'**. As shown in FIG. **37C**, the outer layer (e.g., foxing) **670** can be applied over the housing piece **632'**, in a manner similar to that discussed in connection with FIG. **35A**.

(147) FIG. **38** is a schematic cross-sectional view of the tightening mechanism **608** and concealing portion **614** taken in a plane (e.g., a vertical plane) that intersects the shielding elements (e.g., the side walls **650** and/or **655**). One or both of the side walls **650** and **655** can extend upward at least as far as the sides **624** of the knob **616** in the plane of FIG. **38**, such that the sides **624** of the knob **616** can be partially, mostly, entirely, or substantially entirely covered by the concealing area **614** (similar to the discussion above, e.g., of FIGS. **4-6**). In some embodiments, both the side wall **650** of the securing mechanism and the side wall **655** of the housing **632** can extend upward at least as far as to the top of the knob side **624** (e.g., to substantially the same height, as shown on the right side of FIG. **38**). In some embodiments, the side wall **655** of the housing **632** can extend higher than the side wall **650** of the securing mechanism **634** (as shown on the left side of FIG. **38**). In some embodiments, the side wall **655** of the housing **632** can have a flange portion **663** that extends radially outwardly over at least a portion of the side wall **650**. The flange **663** can clamp the foxing layer **670** down against the side wall **650**.

(148) FIG. **39** is a schematic cross-sectional view of the tightening mechanism **608** and concealing portion **614** taken in a plane in which the concealing portion **614** has a reduced height that is lower than in the plane of FIG. **38**. For example, FIG. **39** can be taken in a plane (e.g., a horizontal plane) that intersects the indented portions **657a** and **657b** and/or **651a** and **651b**. One or both of the side walls **650** and **655** can extend upward to a location rearward of the knob **616**, such that the sides **624** of the knob **616** can be partially, mostly, entirely, or substantially entirely exposed from a side direction. The side walls **650** and **655** can extend upward to substantially the same height (as shown on the right side of FIG. **39**). In some embodiments, the side wall **655** of the housing **632** can extend higher than the side wall **650** of the securing mechanism **634** (as shown on the left side of FIG. **39**). The flange portion **663** can clamp the foxing layer **670** down against the indented portions **651a** and **651b** of the side wall **650**, which can prevent the layer **670** from obstructing the reduced height portions of the concealing area **614**. The spacer **676** can have a greater height for the portions in the plane of FIG. **38** than for the portions of the spacer **676** in the plane of FIG. **39**.

(149) FIG. **40** is a schematic cross-sectional view of the tightening mechanism **608** and concealing portion **614** in which the concealing portion **614** can be compressed to allow a user to actuate the knob **616**. For example, the cross-section of FIG. **40** can be taken in a plane (e.g., a horizontal plane) that intersects the indented portions **657a** and **657b** and/or **651a** and **651b**. The configuration shown in FIG. **40** can be similar to, or the same as, the configuration of FIG. **39** in many regards. The spacer **676** can have a height that is greater than the height of the side walls **650** and/or **655**. In the uncompressed state, shown in FIG. **40**, the concealing portion **614** can extend upward at least as far as the sides **624** of the knob **616** such that the sides **624** of the knob **616** can be partially, mostly, entirely, or substantially entirely covered by the concealing area **614** (similar to the discussion above, e.g., of FIGS. **4-6**). The spacer **676** material can be a compressible so that the concealing portion **614** can be compressed to a compressed state (not shown). In the compressed state, the concealing portion **614** can have a reduced height similar to that shown and discussed in connection with FIG. **39**, such that the user can actuate the knob **616**. The left side of FIG. **40** shows a configuration in which the side wall **655** includes a flange **663**, as discussed above, and the right side of FIG. **40** shows a configuration that does not include the flange **663**. In some embodiments, the compressible areas of the concealing portion **614** can extend around the tightening mechanism **608** by a full 360 degrees, instead of having a portion with rigid shield elements (as shown in FIG. **38**).

(150) FIG. **41** is an exploded view of an example implementation of a tightening mechanism **708** and a concealing portion **714**, which can be used in connection with various embodiments disclosed herein. FIG. **42** shows the assembled tightening mechanism **708** and concealing portion

714. FIG. **43** is a side view of the tightening mechanism and concealing portion **714**. The tightening mechanism **708** can include a housing **732**, a spool **736**, and a knob **716**, which can have features similar to, or the same as the housing **432**, spool **436**, and knob **416** described above. A shaping member **701** can be disposed over the housing **732** to conceal and/or protect the tightening mechanism **708** (e.g., to protect the knob **716**) as discussed herein. The shaping member **701** can be shaped according to the size and shape of the article (e.g., a heel of a shoe) to integrate the tightening mechanism **708** into the appearance of the article. In some embodiments, an outer material (e.g., a foxing) can be disposed over the shaping member **701**, such that the shaping member **701** acts as a spacer to elevate the outer material as discussed herein. In some embodiments, the shaping member **701** can be rigid and can be configured to engage with the housing **732** to position the shaping member **701** and housing **732** at appropriate locations on the article. In some embodiments, an air gap can be formed under the shaping member **701**, e.g., between shaping member **701** and the housing **732**. In some embodiments, the shaping member **701** can be flexible or somewhat flexible, e.g., to allow the shaping member **701** to conform to the particular contours of an article. A supporting member **703** can be disposed between the housing **732** and the shaping member **701**, in some embodiments, to provide support to the shaping member **701** (e.g., to maintain the shape of a flexible shaping member **701**). In some embodiments, the supporting member **703** can be omitted. In some embodiments, the shaping member **701** can include one or more cutouts **705a** and **705b** (e.g., slits) to facilitate bending of the shaping member **701** to conform to the shape of the article. In some embodiments, the shaping member **701** and/or the supporting member **703** can be configured to conceal and/or protect the tightening mechanism **708** more at some locations than at other locations surrounding the tightening mechanism **708**, as discussed herein. The concealing portion **714** can have recesses, cutouts, or scalloped areas, etc. that can provide open portions where the side of the knob **716** is exposed, thereby allowing a user to actuate the knob **716**, as discussed herein.

(151) FIG. **44** is a side view of a shoe having a tightening mechanism **808** and a concealing portion **814** at least partially surrounding the tightening mechanism **808**. In some embodiments, the tightening mechanism **808** can be similar to the tightening mechanism **708** discussed above, although other embodiments disclosed herein can also relate thereto. FIG. **45** shows a shaping member **801**, which can be similar to the shaping member **701** discussed above, with a housing **832** of the tightening mechanism **808** mounted thereto. The knob **816** is not shown in FIG. **45**. FIG. **46** is a cross-sectional view of the shoe of FIG. **44** showing the housing **832** coupled to the shoe and the concealing portion **814**. As discussed in connection with various embodiments herein, the concealing portion **814** can provide areas (e.g., on the sides) in which the tightening mechanism **808** is exposed sufficiently to allow a user to actuate the tightening mechanism **808**.

(152) FIG. **47** is a side view of a shoe having a tightening mechanism **908** and a concealing portion **914** at least partially surrounding the tightening mechanism **908**. FIG. **48** shows another view of the shoe of FIG. **47**. FIG. **49** shows a spacer **976**, which can be configured to provide the shape of the concealing portion **914** of FIGS. **47** and **48**. As discussed in connection with various embodiments herein, the concealing portion **914** can provide areas (e.g., on the sides) in which the tightening mechanism **908** is exposed sufficiently to allow a user to actuate the tightening mechanism **908**.

(153) Although many embodiments are discussed in connection with a tightening mechanism mounted onto the heel of a shoe or other footwear, many other configurations are possible. FIG. **50** is an isometric view of a boot **1000** having a tightening mechanism **1008** mounted onto the tongue **1009** of the boot **1000** and a concealing portion **1014** at least partially surrounding the tightening mechanism **1008**. FIG. **51** is a side view of the boot **1000**. FIG. **52** is a detailed view of the concealing portion **1014** and tightening mechanism **1008** on the boot **1000**. FIG. **53** shows a user actuating the tightening mechanism **1008** of the boot **1000**. Similar configurations are possible for shoes (including high-top shoes and low-top shoes) and other footwear having a tongue. Also, the tightening mechanism **1008** can be mounted onto other portions of the footwear (e.g., on the side

thereof).

(154) As mentioned above, the embodiments described herein can be applied to various articles. For example, FIG. 54 shows a wrist brace **1100** having a tightening mechanism **1108** and a concealing portion **1114** at least partially surrounding the tightening mechanism **1108**.

(155) FIGS. 55a-c show a body or housing **1210** of a tightening mechanism being coupled with a compressible material **1230**, such as a foam backing material. The backing material could be foam of various densities and of materials such as polyurethane or latex rubber, or a non-foam but compliant material such as a polymer gel. The combination of the three parts shown in FIG. 55b is typically coupled to a shoe upper after assembly but before lasting in various potential sequences of assembly and using various assembly methods. Specifically, the body or housing **1210** (hereinafter housing) may be coupled with a foam backing **1230** and then affixed to the rear of a shoe typically with adhesive or by stitching or by RF welding. While being affixed, tubing (not shown) previously mounted between upper layers, may be plugged at its end into tube ports on the housing **1210** through which lace is routed from the front of the shoe to the housing **1210**. Various other embodiments do not use tubing and can allow the tube ports of a housing designed for this purpose to penetrate the shoe surface immediately for the lace coming from the housing **1210** which is then routed externally on the shoe and sometimes with intermediate guiding elements. A relatively rigid mounting component or bayonet **1220** (hereinafter bayonet) is typically joined to a textile or molded overlay known in the shoe industry as a foxing **1250**. These components may be joined by stitching, RF welding, insert molding or by other means. This assembly of bayonet **1220** and overlay may then be affixed to the shoe upper and the bayonet **1220** snapped into receiving elements of the housing **1210**. Often in shoe manufacturing, a subsequent step would involve “back part molding” where the textile upper is placed inside a foot shaped form known as a lasting form and is then heated, and then in this machine the fabric may be pulled and or pushed around the heel shape to somewhat thermoform the heel shape into the materials. The rigid bayonet **1220** firmly holds the perimeter of the housing **1210** hole in the foxing **1250** so that it is not pulled away leaving unsightly gaps between housing **1210** and foxing. This is a key purpose of the relatively rigid bayonet **1220** to resist deformation during back part molding of the hole in the foxing while it is being formed and also to create a neat edge banding with minimal gaps to the material of the foxing **1250**.

(156) In some embodiments, the foam backing **1230** may be molded onto or otherwise coupled with the housing **1210** (e.g. adhered with adhesive or insert molded) so that the foam backing **1230** and housing **1210** appear to be a single or integral piece or component. The foam backing **1230** may be used as a transition component between the tightening mechanism and the shoe to hide any visual defects that may result from attaching the tightening mechanism with the shoe. The foam backing **1230** is relatively compliant material that facilitates in masking or hiding the appearance of marks in the shoe from any underlying components of the tightening mechanism. The foam backing **1230** is able to mask the components by conforming to the specific shape and size of the shoe. For example, when relatively rigid backing materials are used and positioned under the surface of the material of the shoe, the edges of the backing material may be visible or the rigid material may cause the shoe's material to buckle or otherwise deform, which can be visually unappealing. The appearance of underlying components within the shoe is commonly known as ghosting. Ghosting is greatly reduced since foam backing **1230** is compliant and able to adapt and conform to the shape and size the shoe. Specifically, the foam backing **1230** may be able to adapt to the shape and size of the heel counter.

(157) The compliant foam backing **1230** is also capable of adapting to various different shapes and sizes of shoes. This adaptability of the foam backing results in a reduction in the number of backing components that must be manufactured, thereby reducing part count. Foam backing **1230** is adaptable to the various shaped and sized shoes by being insertable and compressible between layers of the shoe. Further, the compliance of foam backing **1230** allows the foam backing **1230** to

be easily wrapped around the heel counter or another component of the shoe regardless of the shoes contour, size, or shape. The foam backing **1230** may be matched to an existing profile of a shoe. For example, the foam piece may be formed to match surrounding surfaces of the article of application (e.g., shoe) so as to provide a seamless visually appealing look.

(158) In some embodiments, the foam backing **1230** may have trimmable parts that allow the shape and/or size of the foam backing **1230** to be adjusted to fit the shape and size of the shoe, such as for example, to particularly adapt to smaller shoe sizes with associated shorter distances from sole to shoe collar. In one embodiment, foam backing **1230** may include a plurality of material layers coupled together in a stacked arrangement, similar to the layers of an onion. Each of the layers may be stripped or peeled away so as to reduce the overall thickness of the foam backing **1230** as desired. In another embodiment, the foam backing **1230** may have perforated portions or regions that allow sections of the foam backing **1230** to be cut or torn away as desired to reduce the size of the foam backing. Similarly, the durometer of the foam may be varied to provide a desired compressibility of the foam material. In some embodiments, the durometer of foam backing **1230** may vary between about 10 and 25 Shore A. By adjusting the durometer of the foam, removing sections, and/or stripping or peeling away various layers of the foam backing **1230**, the foam backing **1230** may be adjusted to conform to a specifically designed shoe. In some embodiments, the foam backing **1230** may include a thermoset material to resist permanent deformation when heated and pressured during back part molding.

(159) In another embodiment, a shim may be positioned under the foam backing **1230** to help the foam backing **1230** conform to and/or adapt to different sized and shaped shoes. For example, when a relatively large thickness of foam backing **1230** is needed or otherwise desired, such as when foam backing **1230** is coupled with a large shoe, a shim may be placed under foam backing **1230** to increase the overall thickness of foam backing **1230**. The shim may comprise any shape or size as desired and may be made of a variety of materials, such as urethane, rubber, an elastomer, and the like. In another embodiment, the foam backing **1230** may include multiple pieces of foam or another material and/or may be unattached to bayonet **1220**.

(160) Bayonet **1220** includes a flange positioned partially or fully around the perimeter of bayonet **1220**. The flange allows the bayonet **1220** to be sewn, adhered, or otherwise coupled with the shoe or other apparel. Housing **1210** couples with bayonet **1220** in a relatively rigid manner. In some embodiment, housing **1210** may be removably coupled with bayonet **1220** so that housing **1210** may be removed for replacement, repair, and the like. In one embodiment, housing **1210** and bayonet **1220** may be coupled together by snapping together mating portions of the housing **1210** and bayonet **1220**. In another embodiment, bayonet **1220** may include bosses that snap or otherwise couple with apertures of the housing **1210**, or vice versa. Cleats may also be used to couple housing **1210** with bayonet **1220**; or the bayonet **1220** may be welded (e.g. heat, RF, ultrasonic, and the like), adhered, or coupled with housing **1210** using any method known in the art. Coupling or interlocking of the housing **1210** with bayonet **1220** using any fastening means described herein (e.g., bosses, cleats, mating components, welding, adhesive bonding, and the like), may facilitate in transferring rotational force from the housing **1210** to the bayonet **1220** as the tightening mechanism is operated. Bayonet **1220** may likewise transfer such force to the shoe or apparel. In this manner, the rotational force is not transferred to foam backing **1230**, which rotational force may cause foam backing **1230** to deform (e.g. become oblong and the like) and/or become visible through a top layer of the shoe or apparel.

(161) FIGS. **56a-b** show the housing **1210** of a tightening mechanism being an integral component of a heel counter **1240** of a shoe. FIGS. **56a-b** are similar to FIGS. **55a-c** except that housing **1210** is molded onto the heel counter **1240** so that heel counter **1240** and housing **1210** are essentially a single component or piece. The single piece heel counter **1240** and housing **1210** may be installed in the shoe as a single unit to eliminate the risk of deformation during construction thereof. Various sizes of these may be molded. In another approach the wings of the heel counter are essentially flat

and may be post trimmed via steel rule dies or other method and then pre-thermoformed to an appropriate curvature for the size of shoe intended. In some embodiments, the bayonet **1220** and foam backing **1230** may be fit over and coupled with housing **1210** as described with respect to FIGS. 55a-c. The material of the shoe **1250**, such as padding, foxing, and the like, may be positioned over the heel counter **1240** and housing **1210** to cover these components and/or to provide padding for the shoe. In this manner housing **1210** may be coupled with the shoe and hidden from view. Often this heel counter/housing combination would be sandwiched between shoe inner liner materials and the outer quarters of the shoe.

(162) Referring now to FIGS. 57a-d, in some embodiments, a cover plate **1310** may be positioned over the housing **1210** of the tightening mechanism. The cover plate **1310** may include a dial cover **1320** that is configured to fit over the knob **1212** of the tightening mechanism so as to cover and hide the knob **1212**. In some embodiments, opposing sides of the dial cover **1320** may be opened so that the sides of knob **1212** are exposed to allow a user to operate the knob **1212** to wind lace about a spool (not shown) of the tightening mechanism as described herein. In some embodiments, the cover plate **1310** may be fit over a foam backing **1230** and bayonet **1220** that are coupled with the housing **1210** as described herein. In other embodiment, the foam backing **1230** and/or bayonet **1220** may not be used and the cover plate **1310** may be fit directly over the knob **1212**.

(163) Heel counter **1240** may include bosses **1242** that allow cover plate **1310** to be coupled with heel counter **1240**, such as by inserting screws through apertures **1312** of cover plate **1310** that correspond with bosses **1242**. In other embodiments, cover plate **1310** may be sewn, adhesively bonded, welded (e.g. heat, ultrasonic, and the like), and the like to heel counter **1240**.

(164) The dial cover **1320** may be a relatively resilient or compliant component that allows the cover plate **1320** to be laterally adjusted relative to cover plate **1310**. Stated differently, the dial cover **1320** may be laterally repositioned relative to cover plate **1310** by stretching dial cover **1320** laterally outward. The adjustability of dial cover **1320** with respect to cover plate **1310** may act on the tightening knob of the reel to allow the tightening mechanism (e.g. knob **1220**) to be pulled axially outward relative to the shoe so as to release a tension on the lace and unwind the lace from a spool of the tightening mechanism as described herein. In this manner, the knob **1220** may be rotated to wind the lace about a spool of the tightening mechanism and subsequently pulled axially outward to unwind the lace from the lace as described herein. In some embodiments, the dial cover **1320** may apply an axial pressure to knob **1220** when the knob **1220** is pulled axially outward so that when a user releases knob **1220**, the knob is biased or forced axially inward and able to be rotated to wind the lace about the spool of the tightening mechanism. In another embodiment, knob **1220** may be rotated in a first direction (e.g., clockwise) to wind lace about the spool and may be rotated in a second direction (e.g., counterclockwise) to unwind lace therefrom. In a specific embodiment, rotation of the spool in a second direction by a defined amount (e.g., between 15 and 90 degrees), may release the tension on the lace and allow the lace to be quickly unwound from the spool.

(165) In some embodiments, the dial cover **1320** may have axial clearance for knob **1212** such that the knob may be grasped through side openings in **1312** such that the knob may stay in the axial outward and released position. Then the compliant and overlaid dial cover **1320** may function as a button so that pressing a top surface of the dial cover **1320** axially inward causes the dial cover **1322** to displace axially between a first position, in which the dial cover **1320** is adjacent the outer surface of the shoe, and a second position, in which dial cover **1320** is positioned axially offset from the shoe. Pressing the dial cover **1320** in this manner may also cause the knob **1212** to axially displace between the first and second position in which the lace may either be wound around the tightening mechanisms spool or unwound therefrom as described herein.

(166) In some embodiments, the cover plate **1310** may include one or more channels (not shown) positioned on an interior surface thereof that define lace paths for the lace of the tightening system. The channels on the interior surface of cover plate **1310** may replace tubing (not shown) which is

commonly used to channel and run lace between various regions or areas of the shoe, such as from the heel to the tongue of the shoe. In another embodiment, tubing (not shown) may be integrated with cover plate **1310** such as being coupled (e.g. adhesively bonded, snapped and the like) with an interior or exterior surface of cover plate **1310**. Cover plate **1310** may be made of a durometer in the range of 20 to 50 Shore A to allow it to conform to various shoe shapes and may also include one or more relief cuts or slots that allow the cover plate **1310** to be flexed so as to accommodate and conform to various shaped and sized shoes. Cover plate **1310** may be a relatively hard plastic material, or a relatively soft, resilient, and flexible material.

(167) Referring now to FIG. **58**, illustrated is another embodiment of coupling a housing **1410** with a shoe. Specifically, the housing **1410** may be integrally formed with an outsole **1402** that is subsequently coupled with the upper material **1420** of the shoe. In one embodiment, the housing **1410** may be insert molded with the outsole **1402**. In another embodiment, the housing may be sewn, adhesively bonded, welded, and the like with outsole **1402**. Since housing **1410** is integrally formed with outsole **1402**, the use of other components to couple the housing **1410** with the shoe (e.g. a bayonet and the like) may not be needed. Likewise, the use of a foam backing may not be needed since ghosting and/or other issues may not be as prevalent. In another embodiment, the housing **1410** may be coupled with the midsole of the shoe that is coupled with the upper material **1420** and/or outsole **1402**. In some cases, tubing for routing lace may be plugged into corresponding housing tube ports. In other cases, the lace may be routed through channels and then along the outside surface of the shoe toward the shoe tongue. In other embodiments, the housing **1410** may be stitched, bonded, glued to the upper and an outsole **1402** may be direct injected to surround the housing **1410**.

(168) Referring to FIGS. **59a-b**, illustrated is another embodiment of coupling the housing **1410** to a shoe. Specifically, the housing **1410** may be integrally formed with an outer material **1430** that is subsequently coupled with this shoe, such as heel counter **1404**. Rather than have the quarters of the shoe sides **1431** overlay the heel counter, in this instance the sides are cut away and do not overlap in order to make a lighter and thinner heel form. The housing **1410** may be pre-attached to the foxing or outer material **1430** via sewing, adhesive bonding, molding, and the like. The foam backing may be sandwiched between the housing **1410** and outer material **1430** during this process. Attaching the housing **1410** to the outer material **1430** in this manner may eliminate the need for one or more other components to be used, such as a bayonet, and the like. Attaching the housing **1410** to the outer material **1430** also allows the housing and tightening mechanism to easily conform to the shape and size of the shoe. The outer material **1430** also covers one or more other components of the tightening mechanism, such as tubing **1406** so that these components are hidden from view of the user. The outer material **1430** may include one or more holes (not shown) and/or channels through which the lace is inserted so that the lace may pass from tubing **1406**, which is positioned on the under surface of outer material **1430**, to the tightening mechanism, which is positioned on the outer surface of outer material **1430**.

(169) Referring now to FIGS. **60a-c**, illustrated is another embodiment of coupling a tightening mechanism **1510** with a shoe **1502**. Specifically, a flexible strip of material **1520** may be coupled over the tightening mechanism **1510** to hide a portion of the tightening mechanism **1510** from view of a user and/or for various other functional reasons, such as to define an outer contour of a heel of the shoe or to provide axial pressure to the tightening mechanism **1510**. In one embodiment, the strip of flexible material **1520** may be positioned over tightening mechanism **1510** so that opposing sides of the tightening mechanism **1510** are exposed and able to be grasped and rotated by a user. In some embodiments, the strip of flexible material **1520** may include a resilient material that allows the tightening mechanism **1510** to be pulled axially outward so that lace may be unwound from a spool of the tightening mechanism. The flexible material strip **1520** may apply an axial force to tightening mechanism **1510** to cause the tightening mechanism **1510** to return to a position axially inward relative to the shoe after lace is unwound from the spool of the tightening

mechanism. The flexible material strip **1520** may provide a relatively visual pleasing appearance to the shoe as well as providing any of the functional aspects described herein.

(170) FIG. **61** is a perspective view of a lacing system **2100** used for tightening a sport shoe **2102**. The sport shoe can be a running shoe, a basketball shoe, and ice skating boot, or snow boarding boot, or any other suitable footwear that can be tightened around a wearer's foot. The lacing system **2100** can be used to close or tighten various other articles, such as, for example, a belt, a hat, a glove, snow board bindings, a medical brace, or a bag. The lacing system can include a reel **2104**, a lace **2106**, and one or more lace guides **2108**. In the illustrated embodiment, the reel **2104** can be attached to the tongue **2110** of the shoe. Various other configurations are possible. For example, the reel **2104** can be attached to a side of the sport shoe **2102**, which can be advantageous for shoes in which the shoe sides **2112a-b** are designed to be drawn closely together when tightened leaving only a small portion of the tongue **2110** exposed. The reel **2104** can also be attached to the back of the shoe **6102**, and a portion of the lace **2106** can pass through the shoe **2102** on either side of the wearer's ankle such that the lace **2106** can be engaged with the reel **2104** when back-mounted.

(171) FIG. **62** is a perspective view of a lacing system **2200** that can be similar to the lacing system **2100**, or any other lacing system described herein. The lacing system can include a reel **2204** which can be similar to the reel **2104**, or any other reel described herein. FIG. **63** is an exploded perspective view of the reel **2204**. FIG. **64** is another exploded perspective view of the reel **2204**.

(172) With reference to FIGS. **62** to **64**, the reel **2204** can include a base member **2214**, a spool member **2216**, and a knob member **2218**. The base member can include a housing **2220** and a mounting flange **2222**. The housing **2220** can include a plurality of housing teeth **2224**, which can extend radially inwardly. The housing **2220** can include lace holes **2226a-b** that allow the lace **2206** to enter the housing **2220**.

(173) The spool member **2216** can be disposed within the housing **2220** such that the spool member **2216** is rotatable about an axis **2228** with respect to the housing **2220**. The lace **2206** can be secured to the spool member **2216** such that when the spool member **2216** rotates in a tightening direction (shown by arrow A) the lace **2206** is drawn into the housing **2220** and is wound around the channel **2230** formed in the spool member **2216**, and when the spool member **2216** rotates in a loosening direction (shown by arrow B) the lace **2206** unwinds from the channel **2230** of the spool member **2216** and exits the housing **2220** via the lace holes **2226a-b**. The spool member **2216** can also include spool teeth **2232** formed thereon. It will be understood that the embodiments disclosed herein can be modified such that rotation in the direction shown by arrow B will tighten the lacing system and such that rotation in the direction shown by arrow A will loosen the lacing system.

(174) The knob member **2218** can be attached to the housing **2220** such that the knob member **2218** can rotate about the axis **2228** with respect to the housing **2220**. The knob member **2218** can include knob teeth **2234** that can be configured to mate with the spool teeth **2232** to couple the knob member **2218** to the spool member **2216** such that rotation of the knob member **2218** in the tightening direction causes the spool member **2216** to also rotate in the tightening direction. In some embodiments, the rotation of the knob member **2218** in the loosening direction can also cause the spool member **2216** to rotate in the loosening direction. The knob member **2218** can also include one or more pawls **2236** which can be biased radially outwardly so as to mate with the housing teeth **2224**. The pawls **2236** and housing teeth **2224** can be configured so that the housing teeth **2224** can displace the pawls **2236** radially inwardly when the knob member **2218** is rotated in the tightening direction, thereby allowing the knob member **2218** to rotate in the tightening direction. The pawls **2236** and the housing teeth **2224** can also be configured so that they engage one another when force is applied to twist the knob member **2218** in the loosening direction, thereby preventing the knob member **2218** from rotating in the loosening direction.

(175) Thus, the reel **2204** can provide a one-way tightening system configured to allow the user to rotate the knob member **2218** in the tightening direction, which causes the spool member **2216** to rotate in the tightening direction, which in turn causes the lace **2206** to be drawn into the housing

2220 via the lace holes **2226a-b**. As the lace **2206** is drawn into the housing **2220** the lacing system **2200** can tighten, causing the lace guide **2208** to be drawn in the direction toward the reel **2204** (shown by arrow C in FIG. **62**). Although the lacing system **2200** is shown with a single lace guide **2208**, any other suitable number of lace guides can be used.

(176) In some embodiments, the knob member **2218** can be axially movable along the axis **2228** between a first or engaged position and a second or disengaged position. FIG. **65** is a side view of the reel **2204** showing the knob member **2218** in the disengaged position drawn in normal lines and showing the knob member **2218** in the engaged position outlined in dotted lines. When in the engaged position, the spool teeth **2232** can engage with the knob teeth **2234** to couple the knob member **2218** to the spool member **2216** as described above. Also, when in the engaged position, the pawls **2236** can engage with the housing teeth **2224** to allow the knob member **2218** to rotate in the tightening direction while preventing the knob member **2218** from rotating in the loosening direction, as discussed above.

(177) When in the disengaged position, the knob member **2218** can be positioned axially further away from the base member **2214** by a distance **2238** that is sufficient to cause the knob teeth **2234** to lift away from and disengage the spool teeth **2232** so that the spool member **2216** is decoupled from the knob member **2218** and the spool member **2216** is free to rotate separately from the knob member **2218**. Thus, the lace **2206** can be withdrawn from the housing **2220** as the spool member **2216** rotates in the loosening direction causing the lacing system **2200** to loosen. When in the disengaged position, the pawls **2236** of the knob member **2218** can be lifted away from the housing teeth **2224** such that they disengage and the knob member **2218** is free to rotate in the both the tightening and loosening direction without restriction. In some embodiments, when the knob member **2218** is transitioned to the disengaged position, the knob teeth **2234** disengage from the spool teeth **2232** and the pawls **2236** also disengage from the housing teeth **2224**. In some embodiments, when the knob member **2218** is transitioned to the disengaged position, the knob teeth **2234** disengage from the spool teeth **2232** while the pawls **2236** continue to engage the housing teeth **2224**. In some embodiments, when the knob member **2218** is transitioned to the disengaged position, the knob teeth **2234** continue to engage the spool teeth **2232** but the pawls **2236** disengage from the housing teeth **2224**.

(178) The distance **2238** between the engaged and disengaged positions of the knob member **2318** can be at least about 1 mm and/or no more than about 3 mm, and can be about 2.25 mm in some embodiments, although distances outside these ranges can also be used. In some embodiments, the distance **2238** can be approximately the same, or slightly greater than, the height of the spool teeth **2232**, the height of the knob teeth **2234**, the height of the housing teeth **2224**, and/or the height of the pawls **2236**.

(179) In some embodiments, because the pawls **2236** engage the housing teeth **2224** in a radial direction while the knob member **2218** is movable between the engaged and disengaged positioned in the axial direction, the reel **2204** can be resistant to accidental disengagement. When the knob member is in the engaged position, and a force is applied to attempt to twist the knob member **2218** in the loosening direction, or lace is pulled tightly causing the spool member **2218** to attempt to twist in the loosening direction, the force is applied to the pawls **2236** as they engage the housing teeth **2224**. Because the pawls **2236** are configured to be displaced radially, not axially, substantially none of the force applied to the pawls **2236** is transferred in the axial direction. Therefore, the reel **2204** can resist higher tightening pressure than some reels in which knob pawls engage housing teeth in the axial direction.

(180) FIG. **66** is a perspective view of the base member **2214**. FIG. **67** is a top view of the base member **2214**. FIG. **68** is a bottom view of the base member **2214**. FIG. **69** is a cross sectional view of the base member **2214**. The base member **2214** a mounting flange **2222** which can be mounted onto the outside structure of an article of footwear or other article, or the mounting flange **2222** can be mounted underneath an outer structure of the article so that at least a portion of the mounting

flange **2222** is hidden from view. The mounting flange **2222** can be secured to the article by stitching, or in any other suitable manner such as using an adhesive, or using rivets, etc. The mounting flange **2222** can be contoured to fit a particular portion of the article (e.g., the back of a shoe), or the mounting flange can be flexible to fit a variety of shapes. The mounting flange **2222** can extend fully or partially around the circumference of the housing **2220**. The mounting flange **2222** can be somewhat resilient to accommodate the flexing of the article during use. In some embodiments, the mounting flange **2222** can be omitted, and the base member **2214** or housing **2220** can be mounted to the article by a screw or rivet or other fastener. For example, a threaded portion of the base member **2214** or housing **2220** can be threaded into a corresponding threaded connector on the article. In some embodiments, the mounting flange **2222** is connected to the article and the reel **2204** is subsequently attached to the flange **2222**.

(181) The housing **2220** can be attached to, or integrally formed with, the mounting flange **2222** and can extend upward therefrom, as illustrated. The housing **2220** can include an outer wall **2240** that surrounds a depression **2242**, which can be substantially circular in shape. A shaft **2244** can extend axially upwardly from the base of the depression **2242**, and the shaft **2244** can be aligned substantially coaxially with the depression **2242**. The shaft **2244** can include a step **2245** or beveled portion where the shaft **2244** meets the base of the depression **2242**. The shaft **2244** can include a bore **2246** in the center thereof which can facilitate the securing of the knob member **2218** to the housing **2220**. The bore **2246** can be threaded or otherwise configured to axially secure a fastener that is inserted therein. The shaft **2244** can form a supporting surface about which the spool member **2216** can rotate.

(182) The outer wall **2240** of the housing **2220** can be substantially cylindrical in shape and can be substantially coaxial with the shaft **2244**. The inner surface of the outer wall **2240** can include a lower portion **2248**, and an upper portion **2250**. The lower portion **2248** can be generally smooth and can include a step **2251** or beveled portion where the outer wall **2240** meets the base of the depression **2242**. The lower portion **2248** can include one or more lace openings **2252a-b** which can be in connected to the lace holes **2226a-b** by lace channels **2254a-b** so that the lace **2206** can pass through the housing **2220** and enter the depression **2242**. As can best be seen in FIG. 69, a lower portion of the lace channels **2254a-b** nearest to the lace holes **2226a-b** can be closed while an upper portion of the lace channels **2254a-b** nearest to the lace openings **2252a-b** can be open at the top. Also, the lace channels **2254a-b** and/or the lace openings **2252a-b** can be in connected to openings **2256a-b** formed in the base of the housing **2220**. The openings **2256a-b** and the open tops of the lace channels **2254a-b** can provide access to the lace **2206** during use and installation, and can also provide an exit pathway for water or other material that may enter the depression **2242** during use, and can facilitate the molding of the lace channels **2254a-b** when the base member **2214** is made of few components (e.g., a single integrated piece).

(183) The housing **2220** can include housing teeth **2224** that extend radially inwardly from the upper portion **2250** of the outer wall **2240**. In the illustrated embodiment, the housing includes 36 housing teeth **2224**, but any other suitable number of housing teeth **2224** can be used. As can best be seen in FIG. 67, each of the housing teeth **2224** can include a first side **2258** and a second side **2260**. The first side **2258** can be shorter than the second side **2260**, and in some embodiments, the first side **2258** can be about half as long as the second side **2260**. In some embodiments, the first side **2258** of the housing teeth **2224** can be at least about 0.5 mm long and/or no more than about 1.0 mm long, and can be about 0.85 mm long, and the second side can be at least about 1.0 mm long and/or no more than about 2.0 mm long, and can be about 1.75 mm long. Other dimensions outside of these specific ranges are also possible. The first side **2258** of the housing teeth **2224** can be angled away from a line that points directly radially inwardly by an angle **2262** that can be at least about 5° and/or at most about 150, and can be about 100 in some embodiments. The second side **2260** of the housing teeth **2224** can be angled away from a line that points directly radially inwardly by an angle **2264** that can be at least about 450 and/or no more than about 650, and can be

about 550 in some embodiments. Other angles outside these specially identified ranges are also possible. In some embodiments, the transition between housing teeth **2224** and between the first and second sides **2258**, **2260** of the housing teeth **2224** can be curved, but hard edged transitions can also be used. The housing teeth **2224** can be configured to interface with the pawls **2236** as discussed in greater detail below. The housing teeth **2224** can include angled top surfaces **2266** to facilitate the transition of the pawls **2236** from the disengaged to engaged positions as will be described in greater detail below.

(184) The base member **2214** can include one or more guard pieces **2268** that can extend axially upwardly further than the outer wall **2240** of the housing **2220** such that the guard piece **2268** can function to cover a portion of the knob member **2218** when the knob member **2218** is attached to the housing **2220**. In some embodiments, the guard piece **2268** can be omitted. In some embodiments, the reel **2204** can be disposed within a recess of the article such that a portion of the article itself extends to cover a portion of the knob member **2218**. The guard **2268**, or portion of the article functioning as a guard, can protect the knob member **2218** and can reduce the occurrence of accidental disengagement of the knob member **2218**.

(185) FIG. **70A** is a perspective view of the spool member **2216**. FIG. **71** is another perspective view of the spool member **2216**. FIG. **72** is a side view of the spool member **2216**. FIGS. **73A-B** are a cross sectional bottom views of the spool member **2216** with the lace **2206** attached thereto. FIG. **74** is a top view of the spool member **2216** disposed within the housing **2220**.

(186) The spool member **2216** can include an upper flange **2270** and a lower flange **2272** with a substantially cylindrical wall **2274** formed therebetween. The outer surface of the wall **2274**, the bottom surface of the upper flange **2270**, and the top surface of the lower flange **2272** can form a channel **2230** for collecting the lace **2206** as it is wound around the spool member **2216**. The inner surface of the wall **2274** can surround a depression **2276** formed in the bottom of the spool member **2216**. A central opening **2278** can extend through the ceiling of the depression. As can best be seen in FIG. **74**, when the spool member **2216** is disposed within the depression **2242** of the housing **2220**, the shaft **2244** can pass through the central opening **2278** of the spool member **2216**. The step **2245** or beveled edge at the bottom of the shaft **2244** can be received into the depression **2276** formed in the bottom of the spool member **2216**. The lower flange **2272** can be formed slightly smaller than the upper flange **2270** (as can best be seen in FIG. **72**) so that the lower flange **2272** can fit inside the step **2251** or beveled edge at the edge of the depression **2242**, and to facilitate removal and/or installation of the spool member **2216** from/into the housing **2220** with the lace **2206** attached. Thus, in some embodiments, the bottom surface of the lower flange **2272** can sit flush against the base of the depression **2242**. In some embodiments, a portion of the housing **2220** can be configured to contact a portion of the spool member **2216** to maintain the bottom surface of the lower flange **2272** a small distance from the base of the depression to reduce the amount of friction as the spool member **2216** rotates. When the spool member **2216** is fully inserted into the depression **2242** of the housing **2220**, the top surface of the upper flange **2270** can substantially align with the top of the lower portion **2248** of the outer wall **2240** such that the upper flange **2270** does not overlap the housing teeth **2224**.

(187) Spool teeth **2232** can be formed on the top surface of the spool member **2216**. In the illustrated embodiment, 12 spool teeth **2232** are shown, but any other suitable number of spool teeth **2232** can be used. Each of the spool teeth **2232** can include a first side **2280** and a second side **2282**. The first side **2280** can be substantially vertical in some embodiments. In some embodiments, the first side can be angled by at least about 50 and/or by no more than about 15°, and in some embodiments by about 100 from the vertical plane. The second side **2282** can be angled by at least about 350 and/or by no more than about 550, and in some embodiments by about 450 from the vertical plane. The first side **2280** can be at least about 1.5 mm long and/or no more than about 2.5 mm long, and can be about 2.0 mm long. The second side can be at least about 2.5 mm long and/or no more than about 3.5 mm long, and can be about 3.0 mm long. Dimensions and

angles outside the identified ranges can also be used. The spool teeth **2232** can be configured to interface with the knob teeth **2234** as discussed in greater detail herein.

(188) In some embodiments, one or more cutouts **2281a-b** can be formed in the upper flange **2270** of the spool member **2216**. Also, in some embodiments, the upper flange **2270** and/or the lower flange can be substantially circular in shape, but can have one or more flattened edges **2283a-d**. The cutouts **2281a-b** and/or the flattened edges **2283a-d** can facilitate the removal of the spool member **2216** from the housing **2220** (e.g., when replacing the lace **2206**). A screwdriver or other tool can be inserted between the spool member **2216** and the housing **2220** wall and the spool member **2216** can be pried out of the housing **2220**. Many variations are possible. For example, FIG. **70B** is a perspective view of a spool member **2216'** which is similar to the spool member **2216** in many respects, except that the upper flange **2270'** and the lower flange **2272'** of the spool member **2216'** do not have flattened edges **2283a-d**. Thus, the upper flange **2270'** and the lower flange **2272'** can be substantially circular in shape. In some embodiments, the upper flange **2270'** can include cutouts **2281a'** and **2281b'** which can facilitate the removal of the spool member **2216'** from the housing **2220**. In some embodiments, the flanges **2270'** and **2272'** that do not include flattened edges **2283a-d** can prevent the lace **2206** from becoming trapped or wedged in the gaps formed between the housing **2220** and the flattened edges **2283a-d**, especially when a relatively thin lace is used.

(189) The depth of the channel **2230** can be at least about 1.5 mm and/or no more than about 2.5 mm, and in some cases can be about 2.0 mm. The channel **2230** can have a width that is at least about 3.0 mm and/or no more than about 4.0 mm, and in some cases can be about 3.5 mm. The outer surface of the wall **2274** can have a diameter of at least about 10 mm and/or no more than about 20 mm, and can be in some cases about 14 mm. Dimensions outside the given ranges are also possible. The lace **2206** can be generally small enough in diameter that the channel **2230** can hold at least about 300 mm of lace and/or no more than about 600 mm of lace, and in some embodiments about 450 mm of lace, although the spool member **2216** and lace **2206** can be configured to hold amounts of lace outside these given ranges.

(190) The lace or cable can have a diameter of at least about 0.5 mm and/or no more than about 1.5 mm, and in some embodiments the diameter can be about 0.75 mm or 1.0 mm, although diameters outside these ranges can also be used. The lace **2206** can be a highly lubricious cable or fiber having a low modulus of elasticity and a high tensile strength. In some embodiments, the cable can have multiple strands of material woven together. While any suitable lace can be used, some embodiments can utilize a lace formed from extended chain, high modulus polyethylene fibers. One example of a suitable lace material is sold under the trade name SPECTRA™, manufactured by Honeywell of Morris Township, N.J. The extended chain, high modulus polyethylene fibers advantageously have a high strength to weight ratio, are cut resistant, and have very low elasticity. One preferred lace made of this material is tightly woven. The tight weave provides added stiffness to the completed lace. The additional stiffness provided by the weave offers enhanced pushability, such that the lace is easily threaded (e.g., into the reel **2204**). Additionally, in some embodiments, the lace can be formed from a molded monofilament polymer. In some embodiments, the lace can be made from woven steel with or without a polymer or other lubrication coating.

(191) One or more ends of the lace **2206** can be secured to the spool member **2216**. In some embodiments, the lace **2206** can be removably or fixedly attached to the spool member **2216**. In some embodiments, the lace **2206** can be threaded through a hole formed in the spool member **2216** and a knot can be formed in the end of the lace **2206**, or an anchoring member can be attached thereto, to prevent the end from being pulled back through the hole. In some embodiments, the lace **2206** can be tied to a portion of the spool member **2216**. The lace can also be secured to the spool member **2216** by an adhesive any other suitable manner. In some embodiments, the lace **2206** is secured to the spool member **2216** by weaving the lace **2206** through a series of openings that cause the lace **2206** to turn at such angles so as to produce

sufficient friction to prevent the lace **2206** from being dislodged from the spool member **2216**. In some embodiments, the lace **2206** wraps over itself so that the lace **2206** tightens on itself when pulled. In some embodiments, only one end of the lace **2206** is secured to the spool member **2216**, with the other end of the lace **2206** being secured to the base member **2214** or to the article being tightened.

(192) The spool member **2216** can include a first set of lace holes **2284a**, **2286a**, **2288a** which can be configured to secure a first end of the lace **2206**. In some embodiments, a second set of lace holes **2284b**, **2286b**, **2288b** can be used to secure the second end of the lace **2206**. Lace guides **2290a-b** can also be formed in the depression **2276** to facilitate the securing of the lace **2206** to the spool member **2216**.

(193) In the embodiment shown in FIG. 73A, a first end of the lace **2206** can pass through the lace hole **2284a** into the depression **2276**. The lace guide **2290a** can direct the lace **2206** toward the lace hole **2286a**, and in some embodiments, the lace guide **2290a** can be positioned such that the lace **2206** is wedged between the lace guide **2290a** and a portion **2292a** of the wall **2274** between the holes **2284a** and **2286a**. The lace **2206** can exit the depression **2276** through the lace hole **2286a** and then turn an angle of approximately 180° to reenter the depression through the lace hole **2288a**. In some embodiments, the tip of the first end of the lace **2206** can be tucked into the opposing lace guide **2290b** to prevent the tip from moving about within the depression **2276** and interfering with the rotation of the spool member **2216**. In some embodiments, the amount of lace **2206** that passes through the lace holes **2284a**, **2286a**, **2288a** can be configured so that only a small portion of the lace **2206** reenters the depression **2276** through the hole **2288a** so that the tip is not tucked into the opposing lace guide **2290b**. The second end of the lace **2206** can be secured to the spool member **2216** by the lace holes **2284a**, **2286b**, **2288b**, and the lace guide **2290b**, and the portion **2292b** of the wall **2274** in like manner.

(194) Other lace securing configurations are possible. For example, in the embodiment shown in FIG. 73B, the first end of the lace **2206** passes through the lace hole **2284a** to enter the depression **2276**. The lace guide **2290** can direct the lace **2206** toward the lace hole **2288b**, and the lace guide **2290a** can be configured such that the lace **2206** is wedged between the lace guide **2290a** and the portion **2294a** of the wall adjacent to the lace hole **2284a**. The lace **2206** can pass through the lace hole **2288b** and then turn an angle of approximately 180° to reenter the depression **2276** through the lace hole **2286b**. The second end of the lace **2206** can be secured to the spool member **2216** by the lace holes **2284b**, **2288a**, **2286a**, and the lace guide **2290b** and the portion **2294b** of the wall **2274** in like manner.

(195) FIGS. 73C and 73D illustrate another manner in which the lace **2206** can be secured to the spool member **2216**. As shown in FIG. 73C, the end of the lace **2216** is threaded through the lace hole **2284a** into the depression **2276**, then through the lace hole **2286a** out of the depression **2276**, and then through the lace hole **2288a** back into the depression **2276**. The end of the lace **2206** can then be passed through the loop in the lace formed between the lace holes **2284a**, **2286a**, as shown in FIG. 73C. The lace **2206** can then be tightened so that the lace crosses under itself as shown in FIG. 73D. For example, the loose end of the lace **2206** can be held with one hand while pulling the loop formed between the lace holes **2284a** and **2286a** to remove the slack from the loop formed between the lace holes **2286a** and **2288a**. Then the slack in the loop formed between the lace holes **2284a** and **2286a** can be pulled out of the depression **2276** through the lace hole **2284a** until the lace tightens down on itself. Thus, once tightened, the lace **2206** bears down on itself more tightly when it is pulled, thereby preventing the lace **2206** from disengaging from the spool member **2216**.

(196) The lace can pass over the top of the portion of the loop that is closest to the lace hole **2288a** and then under the portion of the loop that is furthest from the lace hole **2288a**, as shown. Then, when the lace is tightened, the loose end of the lace **2206** can be directed generally toward the base of the depression **2276**, rather than being directed generally out from the depression **2276** as would be the case if the lace were threaded over the top of the portion of the loop furthest from the lace

hole **2288a**. By biasing the loose end of the lace toward the base of the depression **2276**, the loose end of the lace can be prevented from interfering with the insertion of the spool member **2216** into the housing **2220**. The lace guide **2190a** can be positioned to keep the loose end of the lace **2206** positioned near the periphery of the depression **2276** so that the loose end of the lace **2206** does not enter the central opening **2278** or otherwise interfere with the spool member **2216** being inserted into the housing **2220**.

(197) FIG. **75** is an exploded perspective view of the knob member **2218**. FIG. **76** is another exploded perspective view of the knob member **2218**. The knob member can include a knob core **2296**, pawls **2236**, a spring bushing **2298**, a fastener **2300**, a knob spring **2302**, a knob cover **2304**, and a knob grip **2306**.

(198) The knob core **2296** can be generally disc-shaped. The knob core **2296** can include knob teeth **2234** formed on the bottom surface thereof. In the illustrated embodiment, 12 knob teeth **2234** are shown, but any other suitable number of knob teeth **2234** can be used. In some embodiments, the same number of knob teeth **2234** and spool teeth **2232** can be used, and the knob teeth **2234** can be shaped similar to, or the same as, the spool teeth **2232**, except that that the knob teeth **2234** are oriented in the opposite direction so that the knob teeth **2234** can engage the spool teeth **2232**. Accordingly, the dimensions described above in connection with the spool teeth **2232** can also apply to the knob teeth **2234**. When the knob member **2218** is rotated in the tightening direction, the first sides **2308** of the knob teeth **2234** can press against the first sides **2280** of the spool teeth **2232** to drive the spool member **2216** in the tightening direction. When a lace **2206** is tightened around the spool member **2216** applying a force to the spool member **2216** to cause it to tend to twist in the loosening direction, the second sides **2282** of the spool teeth **2232** can bear against the second sides **2310** of the knob teeth **2234** so that the force is transferred to the knob member **2218** to cause it to tend to twist in the loosening direction. As will be discussed below, the force can cause the pawls **2236** to engage with the housing teeth **2224** to prevent the knob member **2218** and the spool member **2216** from rotating in the loosening direction, thereby maintaining the lace **2206** in the tightened configuration.

(199) The knob core **2296** can include features to facilitate the securing of the knob cover **2304** thereto. The knob core **2296** can include notches **2312** formed in the top surface thereof near the periphery of the knob core **2296**. Protrusions **2314** can extend radially outwardly from the periphery of the knob core **2296** at locations below the notches **2312**. The knob core **2296** can include a central opening **2316** through the center thereof, which can be configured to accept the spring bushing **2298**. A top portion of the central opening **2316** can be wider than a lower portion of the central opening **2316** forming a step **2318** therein. The knob core **2296** can also include features to facilitate the securing of the knob spring thereto, including, for example, a wide engagement tab **2320** and a narrow engagement tab **2322**.

(200) The knob core **2296** can also include pawl depressions **2324**, configured to accept the corresponding pawls **2236**. The pawl depressions **2324** can be generally shaped similarly to the pawls **2236**, but can be somewhat larger than the pawls **2236** to allow the pawls **2236** to pivot and move within the pawl depressions **2324** during operation, as is described in greater detail elsewhere herein. The pawl depressions **2324** can include pawl openings **2326** formed in a portion of the base and/or side thereof to allow a portion of the pawls (e.g., the pawl teeth) to extend through the knob core **2296** (as can be seen in the assembled knob member **2218** shown in FIG. **64**) and interface with the housing teeth **2224**.

(201) FIGS. **77** and **78** are perspective views of a pawl **2236**. The pawl **2236** can include a pawl base **2328**, a pawl beam **2330**, and a pawl spring **2332**. The pawl base **2328** can be configured to interface with the knob core **2296** and/or the knob cover **2304** so that the pawl **2236** can pivot about an axis **2334**. A pivot tab **2336** can extend upward from the pawl base **2328** along the axis **2334**. The pivot tab **2336** can be substantially cylindrical in shape and can be coaxial with the axis **2334**. A flange **2337** can extend out from one side of the pawl base **2328**, and the flange **2337** can

facilitate the pivoting of the pawl **2236**. As can be seen in FIGS. **77** and **78**, in some embodiments, the pawl beam **2330**, the pawl spring **2332**, and other components of the pawl **2236** can be integrally formed (e.g., molded) as a single piece.

(202) The pawl beam **2330** can be formed of a material, thickness, and length such that the pawl beam **2330** is substantially rigid and does not flex as the pawl **2236** is displaced by the housing teeth **2224** when the knob member **2218** is rotated in the tightening direction. One or more pawl teeth **2338a-b** can be positioned near the end of the pawl beam **2330** opposite the pawl base **2328**. In the embodiment shown, two pawl teeth **2338a-b** are used, but any other suitable number of pawl teeth **2338a-b** can be used instead. The pawl teeth **2338a-b**, and in some cases the entire pawl beam **2330**, can have an angled or beveled bottom surface **2339** which can facilitate the transition of the knob member **2218** from the disengaged position to the engaged position, as is discussed in greater detail elsewhere herein. The pawl beam **2330** can include a step **2340** formed where the end of the pawl beam **2330** extends lower than the rest of the pawl **2236**. The downward extending portion of the pawl beam can be configured to extend through, or into, the pawl opening **2326** formed in the pawl depression **2324** of the knob core **2296**.

(203) The pawl base **2328** can include an end surface **2328a** configured to engage surface **2324a** of pawl depression **2324** (as can be seen in FIG. **79**). In some embodiments, as pressure is applied to one or more pawl teeth **2338**, the load can be transferred through pawl beam **2330** to the engagement of end surface **2328a** and surface **2324a**. In some embodiments, as the pawl **2236** pivots radially outwardly about the axis **2334**, the end surface **2328a** of the pawl base **2328** can abut against the surface **2324a** of the pawl depression **2324**, thereby limiting the distance that the pawl **2236** can pivot radially outwardly. For example, the pawl **2236** can be permitted to pivot radially outwardly enough to engage the housing teeth **2224**, but not significantly further. This can relieve pressure off of the pawls **2236** when a loosening force is applied to the knob member **2218**, which can produce a component of force urging the pawls **2236** radially outward, as discussed below. The interface between the surfaces **2328a** and **2324a** can also limit the radial movement of the pawls **2236** when the knob member **2218** is in the disengaged position, thereby keeping the pawls **2236** radially inward enough that the knob member **2218** can be pressed to the engaged position without substantial interference from the pawls **2236**. In some embodiments, pawl **2236** is positioned in pawl depression **2324** and is generally trapped between the knob cover **2304** and the knob core **2296**. As explained below, top tabs **2384** can engage pivot tab **2336** to inhibit axial movement of the pawl **2236**. Similarly, beam tabs **2385** extending downward from knob cover **2304** can engage the upper surface of the pawl beam **2330** to inhibit axial movement thereof.

(204) The pawl spring **2332** can be a cantilever or arch spring as shown in the illustrated embodiment, but any other suitable type of spring can be used. The pawl spring **2332** can extend out from the pawl base **2328** in the same general direction as the pawl beam **2330**. The pawl spring **2332** can be curved away from the pawl beam **2330**. A generally cylindrically shaped end piece **2342** can be formed at the end of the pawl spring. The pawl spring **2332** can be made of a material, thickness, and length such that the pawl spring **2332** is resiliently flexible so that it flexes as the pawl **2236** is displaced by the housing teeth **2224** when the knob member **2218** is rotated in the tightening direction. The pawl spring **2332** as shown in the relaxed position in FIGS. **77** and **78**. In some embodiments, the pawl beam **2330** and the pawl spring **2332** are independently formed and then coupled to form the pawl **2236**. Thus, pawl beam **2330** and pawl spring **2332** need not be formed of the same material. For example, a metal pawl beam **2330** may be advantageous because of its relatively high strength to thickness ratio while it may be advantageous to use a plastic pawl spring **2332**. In some embodiments, the same material may be used in each, even when the beam pawl beam **2330** and the pawl spring **2332** are separately formed. In the illustrated embodiment of FIGS. **77-78**, the pawl spring **2332** and the pawl beam **2330** can be integrally formed of the same material as a single piece, thereby simplifying the manufacturing and assembly cost and complexity. In some embodiments, different springs may be used than that shown in the illustrated

embodiments. For example, a metal or plastic leaf spring or a wire coiled spring may be used in some applications.

(205) Because the pawl beam **2330** and pawl spring **2332** are separate portions, the pawl spring **2332** can be altered to be more easily flexible (e.g., by making the pawl spring **2332** thinner) without reducing the amount of force the pawl beam **2330** is able to withstand as the knob member **2218** is twisted in the loosening direction. Likewise, the pawl beam **2330** can be altered so that it can withstand greater force applied to the knob **2218** in the loosening direction (e.g., by making the pawl beam **2330** thicker) without making the pawl spring **2332** less flexible. Thus, the pawl **2236** can be tuned to a desired level of flexibility and strength. For example, a pawl **2236** can be configured to withstand large amounts of force when the knob member **2218** is twisted in the loosening direction while also being easily radially displaceable when the knob member **2218** is rotated in the tightening direction. In some embodiments, the force applied to the pawl **2236** when the knob member **2218** is twisted in the loosening direction is born by the pawl beam **2330** and substantially none of the force is born by the pawl spring **2332**. This configuration can be advantageous over embodiments in which a pawl includes a load bearing beam that also flexes to displace the pawl (e.g., during tightening), because the load bearing capability of the flexible pawl is reduced as the pawl is made more flexible, and the flexibility of the pawl is reduced as the beam is made to withstand higher forces. Thus, when using the flexible beam pawl, a sufficient amount of loosening force can cause the pawl beam to buckle, thereby compromising the lacing system. However, when using the pawls **2236**, the pawl beam **2330** can be configured to be substantially rigid even when a relatively large loosening force is applied, and the pawl spring **2332** can be configured to allow the pawl beam **2330** to pivot easily when a tightening force is applied.

(206) FIG. **79** is a top view showing the pawls **2236** positioned inside of the pawl depressions **2324** of the knob core **2296**. Although the housing **2220** is not shown in FIG. **79**, the pawls **2236** are shown in the position where the pawl teeth **2338a-b** are engaged with the housing teeth **2224**. FIG. **80** is a top view showing the base member **2214** and the pawls **2236** in the same position as in FIG. **79** with the pawl teeth **2338a-b** engaged with the housing teeth **2224**. FIG. **81** is a top view of the base member **2214** and the pawls **2236** in a displaced configuration as the knob member **2218** is rotated in the tightening direction. The elements of the knob member **2218**, other than the pawls **2236**, and the spool member **2216** are omitted from the view shown in FIGS. **80** and **81** for simplicity.

(207) In some embodiments, the pawl springs **2332** can be partially flexed to a position that is less curved than the relaxed position when inserted into the pawl depressions **2324**. The flexed pawl springs **2332** can cause the pawls **2236** to tend to pivot so that the pawl beams **2330** are biased radially outwardly and so that the pawl teeth **2338a-b** bear radially outwardly against the housing teeth **2224**. When the knob member **2218** is twisted in the loosening direction (shown by arrow B) the first sides **2344a-b** of the pawl teeth **2338a-b** can bear against the first sides **2258** of the housing teeth **2224** to prevent the knob member **2218** from rotating in the loosening direction. In some embodiments, the pawl depressions **2324** can be configured to receive the pawls **2236** without the pawl springs **2332** needing to be partially flexed. Thus, in some embodiments, the pawl springs **2332** can be in the relaxed position when the pawl beams **2330** are engaged with the housing teeth **2224** to prevent the knob **2218** from loosening. When the pawl beams **2330** are displaced away from the housing teeth **2224**, the pawl springs **2332** can transition from a relaxed to a flexed state such that the pawl beams **2330** are biased toward the housing teeth **2224**. Also, as shown for example in FIG. **80**, in some embodiments, one or more of the pawl teeth **2338a-b** can engage the housing teeth **2224** at locations that are radially outside a tangent line that extends from the pivot axis **2334** of the pawl **2236**. In the embodiment of FIG. **80**, the pawl tooth **2338b** can engage the corresponding housing tooth **2224** at a location on a line that is angled radially outward from the tangent line C by an angle **2345** that is at least about 5° and/or less than or equal to about 15°, and can be about 10° in some embodiments. Thus, when a loosening force is applied

to the knob member **2218** (shown by arrow B), a component of the force is directed to urge the pawl **2236** to pivot radially outwardly. Thus, as more loosening force is applied to the knob member **2218**, the pawl teeth **2338a-b** are urged to engage the housing teeth **2224** more firmly. This can prevent the pawls **2236** from unintentionally disengaging from the housing teeth **2224** when a large loosening force is applied. As the pawl **2236** is urged radially outward, the pawl beam can abut against the tips of one or more housing teeth **2224** not engaged by the pawl teeth **2338a-b**, which can prevent the pawl beam **2330** from buckling outwardly and can transfer some of the loosening force into the housing. As discussed above, the surface **2328a** of the pawl base **2328** can abut against the surface **2324a** of the pawl depression **2324**, thereby limiting the amount that the pawl **2236** can rotate radially outwardly.

(208) In some embodiments, multiple pawl teeth **2338a-b** can be used so that the multiple pawl teeth **2338a-b** simultaneously engage multiple corresponding housing teeth **2224** so that, when the knob member **2218** is twisted in the loosening direction, the applied force is distributed across multiple teeth per pawl **2236** to prevent the knob member **2218** from rotating in the loosening direction. By distributing the force across multiple teeth, the housing teeth **2224** and pawl teeth **2338a-b** can be relatively small in size while still providing sufficient engagement surface area between the first sides **2258** of the housing teeth **2224** and the first sides **2344a-b** of the pawl teeth **2338a-b**. For example, the engagement of two pawl teeth **2338a-b** with two consecutive housing teeth **2224** as shown can provide substantially the same engagement surface area for resisting rotation in the loosening direction as a single pawl tooth and housing tooth of twice the size shown. As the size of the housing teeth **2224** is reduced, the number of housing teeth **2224** can increase, and the tightening resolution of the reel **2204** can increase. When the knob member **2218** is advanced by one housing tooth **2224** in the tightening direction (shown by arrow A), the rotational distance that the knob member **2218** travels is reduced as the size of the housing teeth **2224** is reduced and the number of housing teeth **2224** is increased. Thus, by using more, and smaller, housing teeth **2224**, the tightening resolution of the reel **2204** is increased so that the lacing system **2200** can be tightened more precisely to the desired level of tightness. Also, as the size of the housing teeth **2224** is reduced, the distance that the pawls **2236** are displaced in the radially inward direction when the knob member **2218** is tightened is also reduced, thereby making the knob member **2218** easier to rotate in the tightening direction. It is important to note that, in some embodiments, because the multiple pawl teeth **2338a-b** are used, the knob member **2218** can be easily rotated in the tightening direction while strongly resisting rotation in the loosening direction. Although two pawl teeth **2338a-b** are shown per pawl **2236**, additional pawl teeth (e.g., three, four, five, or more) can be used, and, in some embodiments, a single pawl tooth can be used. As shown for example in FIG. **80**, in some embodiments, one or more of the pawl teeth **2338a-b** and the housing teeth **2224** can be configured to lock together when fully engaged, thereby preventing the pawl **2236** from rotating radially inward unless the knob member **2218** is moved in the tightening direction (shown by arrow A). The surface **2258** of the housing tooth **2224** and the surface **2344a** of the pawl tooth **2338a** can be form an angle **2343** (e.g., by at least about 5° and/or by less than or equal to about 15° , or by about 10°) from a line D, which can be perpendicular to the tangent line C for the pivot axis **2334** of the corresponding pawl **2236**. The line D can be tangent to the arc tracked by the surface **2344a** of the pawl tooth **2338a** as it pivots radially inward. Since the surface **2258** of the housing tooth **2224** is angled towards the pawl beam **2330**, the surface **2334a** can abut against the surface **2258** when a force urges the surface **2334a** to move in the direction of arrow D. Thus, when the pawl tooth **2338a** fully engages the housing tooth **2224** such that the surface **2344a** of the pawl tooth **2338a** abuts against the surface **2258** of the housing tooth **2224**, the pawl **2236** is prevented from rotating in the radially inward direction because radially inward rotation would cause the surface **2344a** of the pawl tooth **2338a** to press more firmly against the surface **2258** of the housing tooth **2224**. The angled interface between the surfaces **2258** and **2344a** can also provide a force on the pawl **2236** in the radially outward direction when a loosening force is

applied (shown by arrow B). To allow the pawl **2236** to rotate radially inwardly, the pawl **2236** can be shifted in the tightening direction (shown by arrow A) so that the surface **2344a** of the pawl tooth **2338a** disengages from the surface **2258** of the housing tooth **2224**. The other pawl teeth (e.g., pawl tooth **2338b**) can operate similar to the pawl tooth **2338a** to prevent unintentional disengagement of the pawls **2236**.

(209) When the knob member **2218** is rotated in the tightening direction (shown by arrow A), the second sides **2260** of the housing teeth **2224** can slide along the second sides **2346a-b** of the pawl teeth **2338a-b**, causing the pawls **2236** to rotate about the pivot axis (e.g., about the pivot tab **2336**) so that the pawl beams **2330** are displaced radially inwardly away from the housing teeth **2224**, as shown in FIG. **81**. As the pawls **2236** rotate, the pawl springs **2232** can be further flexed, for example to a position that is less curved, and the end piece **2342** can slide along the wall of the pawl depression **2224** that is further away from the pawl base **2328**. The curved edge of the generally cylindrically shaped end piece **2342** can provide a small contact area between the end piece **2342** and the wall of the pawl depression **2224** to reduce the amount of friction therebetween as the end piece **2342** slides. Once the tips of the pawl teeth **2338a-b** pass the tips of the housing teeth **2224**, the pawls **2236** can snap radially outwardly to a position similar to that shown in FIG. **80** except that the pawls **2236** are advance by one housing tooth **2224**, or one step, in the tightening direction. To tighten the lacing system **2200**, the user can rotate the knob member **2218** in the tightening direction by a desired amount, with the pawls **2236** snapping back after each step to prevent rotation in the loosening direction.

(210) As can be seen in FIGS. **80** and **81**, the flanges **2337** of the pawls **2236** can extend radially outwardly past the tips of the housing teeth **2224**, but the flanges **2337** can be positioned near the tops of the pawls **2236** where the flanges **2337** do not contact the housing teeth **2224**. Rather, the flanges **2337** can contact a portion of the wall **2325** of the pawl depressions **2324**, as can be seen in FIG. **79**. As the pawls **2236** rotate, the flanges **2337** can roll slightly against the wall of the pawl depressions **2324** to facilitate the desired rotational displacement of the pawls **2236**. The mating of flange **2337** and wall portion **2325** can also assist in maintaining the general radial and axial position of the pawl **2236** in the pawl depression **2324**.

(211) The pawls **2236** can be configured differently than as shown in the illustrated embodiments. For example, in some embodiments, the flexible arm of the pawl spring **2332** can curve toward the pawl beam **2330** (e.g., in the opposite direction as that shown in the illustrated embodiments), and a middle portion of the curved arm of the pawl spring **2332** can ride along a wall of the corresponding depression **2324**. In some embodiments, the curved arm can be configured so that it is more curved when in the more flexed position (e.g., when the pawl beam **2330** is displaced away from the housing teeth **2224**) than when in the less flexed position (e.g., when the pawl beam **2330** is engaged with the housing teeth **2224**). In some embodiments, the flexible arm can be attached to the pawl **2236** at locations other than that shown in the illustrated embodiment. For example, the flexible arm of the pawl spring **2332** can be extend from the end of the pawl beam **2330** that is furthest from the pivot tab **2336**. Other variations are possible. Also, in some embodiments, the pawl spring **2332** can include a flexible arm that extends in generally the opposite direction as the pawl beam **2330**, or generally radially inwardly, or in various other suitable directions so long as the pawl spring **2332** can be flexed to bias the pawl beam **2330** toward the housing teeth **2224**. As discussed above, the pawl spring **2332** can also be made from a leaf spring, or a coil spring, or any other suitable biasing member configured to bias the pawl beam **2330** radially toward the housing teeth **2224**.

(212) Although various embodiments discussed herein include housing teeth **2224** that extend radially inwardly and pawls **2236** configured to be biased radially outwardly toward the housing teeth **2224**, other configurations are possible. For example, the housing teeth **2224** can extend radially outwardly. The housing teeth **2224** can be formed, for example, on the outside surface of the shaft **2244** or similar structure. In these embodiments, the pawls **2236** can be configured to be

biased radially inwardly toward the housing teeth **2224**. In some embodiments it may be advantageous to position the housing teeth **2224** nearer to the periphery of the reel **2204** (e.g., as shown in the illustrated embodiments) so that the housing teeth **2224** are disposed along a larger circumference so that more housing teeth **2224** can be included, thereby increasing the tightening resolution (the number of teeth per revolution) of the reel **2204**.

(213) FIG. **82** is a top view of the knob core **2296**, the spring bushing **2298**, the fastener **2300**, and the knob spring **2302** in the assembled configurations. With reference now to FIGS. **75**, **76**, and **82**, the spring bushing **2298** can be generally cylindrical in shape and can have a central opening **2348** formed through the center thereof. The outer surface of the spring bushing **2298** can be wider at a top portion **2349** than at a bottom portion **2351**, forming a step **2350** which can be configured to abut against the step **2318** formed in the central opening **2316** of the knob core **2296** when the spring bushing **2298** is fully inserted into the central opening **2316** of the knob core **2296**. In the central opening **2348** that passes through the center of the spring bushing **2298**, the upper portion can be wider than a lower portion, to form a step **2352**.

(214) The head **2354** of the fastener **2300** can abut against the step **2352** in the central opening of the spring bushing **2298** when the fastener **2300** is fully inserted into the central opening **2348** of the spring bushing **2298**. The fastener **2300** can be a screw having a shaft **2356** that includes threads **2358** configured to engage the threads formed in the bore **2246** formed in the shaft **2244** of the housing. In some embodiments, the bore **2246** can include a threaded metal insert or a plastic thread molded as part of the bore **2246**. In some embodiments, the bore **2246** does not have preformed threads, and the threads **2358** of the fastener **2300** can form the threads in the bore the first time that the fastener **2300** is inserted into the bore **2246**. The head **2354** can include a notch **2360**, which can be hexagonally or cross shaped, or otherwise configured to allow a screwdriver or other tool to turn the fastener **2300**. In some embodiments, the knob member **2218** can be coupled to the housing **2220** in some other way, such as using a snap together fastener or rivet or ultrasonic welding. Other alternatives are possible.

(215) The knob spring **2302** can include a pair of opposing engagement portions **2362a-b** which can be configured to engage the spring bushing **2298**. A pair of end pieces **2364a-b** can extend approximately orthogonally from the engagement portions **2362a-b** in an inward direction. An interconnecting portion **2368**, which can be shaped to follow the partial circumference of a circle, can be attached to the engagement portions **2362a-b** by curved connectors **2370a-b**.

(216) The knob spring **2302** can be secured to the knob core **2296**. The wide engagement tab **2320** can be configured to fit between the curved connectors **2370a-b** of the knob spring **2302**, and the narrow engagement tab **2322** can be configured to fit between the end pieces **2364a-b** of the knob spring **2302** to prevent the knob spring **2302** from rotating or otherwise moving with respect to the knob core **2296**. In some embodiments, the wide engagement tab **2320** and/or the narrow engagement tab **2322** can be configured to receive the knob spring **2302** so that the knob spring **2302** is maintained in a slightly flexed configuration with the curved connectors **2370a-b** bearing against the wide engagement tab **2320** and/or the end pieces **2364a-b** bearing against the narrow engagement tab **2322**. In some embodiments, the knob spring **2302** can be prevented from moving axially by the knob cover **2304** when it is attached to the knob core **2296**.

(217) The knob spring **2302** can be configured such that the engagement portions **2362a-b** can be resiliently moved apart from one other to allow the upper wide portion **2349** of the spring bushing **2298** to pass between the engagement portions **2362a-b**. The spring bushing **2298** can be in a disengaged position, as shown in FIG. **82**, where the spring bushing **2298** is located below the engagement portions **2362a-b**. In the engaged position, the upper wide portion **2349** of the spring bushing **2298** can be disposed above the engagement portions **2362a-b** of the knob spring **2302**. The upper wide portion **2349** of the spring bushing can be wider than the distance between the engagement portions **2362a-b** of the knob spring **2302** to prevent the spring bushing from inadvertently transitioning between the engaged and disengaged positions. To transfer the spring

bushing **2298** from the engaged to the disengaged positions, a force can be applied, for example by pulling the knob member **2218** in the axial direction away from the base member **2214**, that causes the spring bushing **2298** to press down against the engagement portions **2362a-b** causing the engagement portions **2362a-b** to resiliently separate from one another until the upper wide portion **2359** of the spring bushing **2298** passes between the engagement portions **2362a-b**. To transfer the spring bushing **2298** from the disengaged to the engaged positions, a force can be applied, for example by pushing the knob member **2218** in the axial direction toward the base member **2214**, that causes the spring bushing **2298** to press up against the engagement portions **2362a-b** causing the engagement portions **2362a-b** to resiliently separate from one another until the upper wide portion **2359** of the spring bushing **2298** passes between the engagement portions **2362a-b**.

(218) Many variations are possible. For example, in some embodiments, the engagement portions **2362a-b** can be maintained rigidly in place and the spring bushing **2298** can be made from a resiliently compressible material so that the spring bushing **2298** can transition between the engaged and disengaged positions by resiliently compressing and passing between the engagement portions **2362a-b**. In some embodiments, the fastener **2300** and the spring bushing **2298** can be combined into a single piece. The knob spring **2302** can assume a variety of other shapes and can be attached to the knob core **2296** in a variety of other manners such that the engagement portions **2262a-b** are configured to resiliently flex away from one another. The spring bushing **2298** can be formed in various other shapes than that shown in the illustrated embodiments. In some embodiments, the spring bushing **2298** can be rotationally asymmetrical and can rotate with the knob core **2296** and knob spring **2302**. Thus, in some cases, the spring bushing **2298** can have flat sides that engage the knob spring **2302** along a line instead of just at a point.

(219) With reference now to FIGS. **75** and **76**, the knob cover **2304** can be generally disc shaped. The knob cover **2304** can have a domed or generally frustoconical top wall **2372** and a peripheral wall **2374** with a cavity **2376** formed therein. A central opening **2378** can be formed at the center of the top wall **2372** to allow a screwdriver or other tool to be inserted therethrough to engage the notch **2360** on the fastener **2300**. The knob cover **2304** can include securing tabs **2380** and notches **2382** configured to engage the corresponding notches **2312** and protrusions **2314** on the knob core **2196** to secure the knob cover **2304** to the knob core **2296** using a snap-fit connection. The knob cover **2304** can be secured to the knob core **2296** in various other ways such as using an adhesive, a threaded connection, ultrasonic welding, or any other suitable manner. The knob cover **2304** can be either fixedly or removably attached to the knob core **2296**. When the knob cover **2304** is attached to the knob core **2296**, the pawls **2236**, the spring bushing **2298**, the fastener **2300**, and the knob spring **2302** can be enclosed therebetween.

(220) Top tabs **2384** can extend downward from the underside of the top wall **2372** of the knob cover **2304**. The top tabs **2384** can align with the pivot tabs **2336** of the pawls **2236**, and the bottom surfaces of the top tabs **2384** can contact, or nearly contact, the top surfaces of the pivot tabs **2336** of the pawls **2236** to thereby prevent the pawls from moving axially. Many variations are possible. In some embodiments, the pivot tabs **2336** of the pawls **2236** can fit into bores formed in the knob cover **2304** to secure the pawls **2236** and allow the pawls **2236** to pivot about the pivot tabs **2336**.

(221) A recess **2386** can be formed at the center of the cavity **2376**, and the recess **2386** can be configured to receive the upper wide portion **2349** of the spring bushing **2298** when the spring bushing **2298** is in the engaged position.

(222) The peripheral wall **2374** of the knob cover **2304** can include notches **2388** configured to receive corresponding tabs **2390** formed on the inside surface of the knob grip **2306**. The knob grip **2306** can be generally doughnut shaped and can include raised portions **2392** and/or depressions **2394** on the outside surface to facilitate the gripping of the knob member **2218**. In some embodiments, the knob grip **2306** can be omitted or can be divided into intermittent portions disposed about the periphery of the knob cover **2304**. Other variations are possible.

(223) An opening **2396** can be formed in a portion of the top wall **2372** of the knob cover **2304** to

provide a view of some of the internal components of the reel **2204** during use, or to provide an exit path for water or other foreign material to exit the reel **2204**. In some embodiments, the opening **2396** can be omitted.

(224) As mentioned above, the knob member **2218** can be axially movable between engaged and disengaged positions. FIG. **83A** is an exploded view of the reel **2204** with the knob member **2218** in the engaged configuration. FIG. **83B** is a cross sectional view of the reel **2204** with the knob member **2218** in the engaged configuration. FIG. **84A** is an exploded view of the reel **2204** with the knob member **2218** in the disengaged configuration. FIG. **84B** is a cross sectional view of the reel **2204** with the knob member **2218** in the disengaged configuration. The knob member **2218** can be secured to the base member **2214** by twisting the fastener **2300** so that the threads **2358** mate with corresponding threads in the bore **2246** formed in the shaft **2244**. In some embodiments, when the fastener **2300** is sufficiently tightened, the portion of the shaft **2244** that extends up past the spool member **2216** can enter into a lower portion of the central opening **2348** formed through the spring bushing **2298**. The bottom edge **2398** of the spring bushing **2298** can abut against, or nearly contact, the annular region **2400** inside of the spool teeth **2232**.

(225) When the knob member **2218** is in the engaged position, as shown in FIGS. **83A** and **83B**, the spring bushing **2298** and the fastener **2300** can be maintained in an raised position by the knob spring **2302**, as discussed above, so that the bottom edge **2398** of the spring bushing **2298** does not extend past the central opening **2316** of the knob core **2296**. Thus, the knob member **2218** is maintained in the lower engaged position (shown in dotted lines in FIG. **65**), with the bottom of the knob core **2296** abutting against, or in close proximity to, the top surface of the spool member **2216**. Thus, when in the engaged position, the knob teeth **2234** engage the spool teeth **2232**, and the pawls **2236** engage the housing teeth **2224**.

(226) When the knob member **2218** is in the disengaged position, as shown in FIGS. **84A** and **84B**, the spring bushing **2298** and the fastener **2300** can be maintained in a lowered position by the knob spring **2302**, as discussed above, so that the bottom edge **2398** of the spring bushing **2298** extends past the central opening **2316** of the knob core **2296** by at least about 1.0 mm and/or by no more than about 3.0 mm, and in some embodiments by about 2.25 mm, although other configurations outside these ranges are also possible. Since the bottom edge **2398** of the spring bushing **2298** continued to abut against, or nearly contact, the annular region **2400** of the spool member **2216**, the knob member **2218** is raised away from the spool member **2216** and base member **2214** by an amount (e.g., about 2.25 mm) sufficient to cause the knob teeth **2234** to disengage from the spool teeth **2232** and/or to cause the pawls **2236** to disengage from the housing teeth **2224**. In the embodiment shown, when the knob is in the disengaged position, the knob teeth **2234** disengage from the spool teeth **2232** and the pawls **2236** also disengage from the housing teeth **2224**. Thus, in the illustrated disengaged configuration the spool member **2216** can be free to rotate in the loosening direction independent of the knob member **2218** to loosen the lacing system **2200**, and the knob member **2218** can be free to rotate in both the tightening and loosening directions.

(227) Many variations are possible. In some embodiments, when in the disengaged position, the knob teeth **2234** can disengage from the spool teeth **2232** while the pawls **2236** continue to engage the housing teeth **2224** (e.g., if the step **2340** shown in FIG. **77** were made larger so that the pawl teeth **2338a-b** extended further downward). In these embodiments, the knob member **2218** can be impeded from rotating in the loosening direction even when in the disengaged position, but the spool member **2216** can be free to rotate in the loosening direction independent of the knob member **2218** to allow the lace **2206** to be withdrawn to loosen the lacing system **2200**. In some embodiments, when in the disengaged position, the knob teeth **2234** can continue to engage the spool teeth **2232** (e.g., if the knob teeth **2234** and/or the spool teeth **2232** were made taller than in the illustrated embodiments) while the pawls **2236** can disengage from the housing teeth **2224**. In these embodiments, the spool member **2216** continues to be coupled to the knob member **2218** even when in the disengaged position, but the knob member **2218** and spool member **2216** are

permitted to rotated together in the loosening direction to release the lace **2206** from the reel **2204** to loosen the lacing system **2200**. Other variations are also possible. For example, in some embodiments, the spool member **2216** can be integrally formed with, or fixedly attached to, or removably attached to the knob member **2218**, and the spool teeth **2232** and knob teeth **2234** can be omitted.

(228) As mentioned above, when in the disengaged position, the pawls **2236** can be raised sufficiently to disengage from the housing teeth **2224**. In some embodiments, because the pawls are biased radially outwardly by the pawl springs **2232**, the pawls **2236** can deflect radially outwardly so that portions of the bottom surfaces of the pawls **2236** are positioned above portions of the top surfaces of the housing teeth **2224**. Thus in some embodiments, when the knob member **2218** is transitioned back to the engaged position, the pawls **2236** must be deflected radially inwardly so that they can reengage with the housing teeth **2224**. As also mentioned above, at least a portion of the top surfaces **2266** of the housing teeth **2224** can be angled or beveled and/or at least a portion of the bottom surfaces **2339** of the pawls **2236** can be angled or beveled, so that the downward pressure applied when the knob member is returned to the engaged position can cause the pawls **2236** to deflect radially inwardly to facilitate the reengagement of the pawls **2236** with the housing teeth **2224**. In some embodiments, the pawl depressions **2324** or other portions of the knob member **2218**, can be configured to prevent the pawls **2236** from deflecting radially outwardly past the radial position where the pawls **2236** engage the housing teeth **2224**, thereby reducing or eliminating the need to deflect the pawls **2236** inwardly when transitioning the knob member **2218** to the engaged position.

(229) The knob member **2218** can be transitioned from the engaged position to the disengaged position by pulling the knob member **2218** axially away from the base member **2214** with enough force to cause the spring bushing **2298** to displace the knob spring **2302** and pass therethrough. To transition the knob member **2218** from the disengaged position to the engaged position the knob member **2218** can be pushed in the axial direction toward the base member **2214** with enough force to cause the spring bushing **2298** to displace the knob spring **2302** and pass therethrough.

(230) The radial engagement of the pawls **2236** with the housing teeth **2224** can reduce or eliminate the occurrence of unintentionally transitioning the knob member **2218** from the engaged to disengaged positions by applying force to tend to twist the knob member **2218** in the loosening direction. If the lace **2206** is pulled, it can impart a force tending to twist the spool member **2216** in the loosening direction, and the force can be transferred to the knob **2218** via the spool teeth **2232** and knob teeth **2234**, and the pawls **2236** can distributed the force radially among a certain number of the housing teeth **2224**. Because the pawls **2236** engage the housing teeth radially, not axially, and because the pawls **2236** are configured to be displaced radially (when tightening the reel **2204**), substantially none of the force is applied to the knob **2218** in the axial direction. Thus, the radial pawls **2236** do not impart any substantial force in the direction of the axial direction that would tend to separate the spool teeth **2232** from the knob teeth **2234** which can lead to unintentional disengagement of the knob member **2218** and/or unintentional loosening of the spool member **2216**. Thus, the reel **2204** can be configured to withstand greater amounts of force applied to pull on the lace **2206** or applied to try and twist the knob member **2218** in the loosening direction without unintentionally causing the knob member **2218** to disengage than a reel **2204** in which the pawls axially engage the housing teeth and the pawls are configured to displace axially during tightening.

(231) Also, in some embodiments, the force applied to the pawls **2236** when the knob **2218** is twisted in the loosening direction is born by the pawl beams **2330** such that substantially none of the force is transferred to the pawl springs **2332**. Thus, the pawl springs **2332** can be configured to be easily flexible while the pawl beams **2330** can be configured to be substantially rigid. Therefore, the pawls **2236** can be configured to resist a relatively large amount of force applied to twist the knob member **2218** in the loosening direction because that force is born by the rigid pawl beams

2330, while the pawls can also be configured to rotate radially when a relatively small force is applied to twist the knob member **2218** in the tightening direction because that force is transferred to the flexible pawl springs **2332**.

(232) The components of the lacing systems described herein can be formed from any suitable material such as, but not limited to, plastic, carbon or other fiber reinforced plastic, aluminum, steel, rubber, or any other suitable material or combination of such materials. In some embodiments, the base member **2214**, spool member **2216**, knob core **2296**, pawls **2236**, spring bushing **2298**, knob cover **2304**, lace guides, or any other suitable components described herein can be injection molded or otherwise formed from any suitable polymeric material, such as nylon, PVC or PET. Some of the components described herein can be formed from a lubricious plastic such as PTFE, or other material useful in reducing the friction between a lace and such components as desired. Additionally, some of the components described herein can be coated or layered with a lubricious material to reduce the friction with interacting components or parts. The fastener **2300**, and the knob spring **2302** can be made from a metal (e.g., aluminum or steel), but other materials can also be used such as plastics. The knob grip **2306** can be formed from rubber, or latex, or silicon, or any other material to facilitate the gripping of the knob member **2218**.

(233) FIG. **85** is a perspective view of an alternative embodiment of a base member **2414** which can be used in place of the base member **2214** discussed above. The base member **2414** can include a housing **2420** and a mounting flange **2422** and can be generally similar to the base member **2214** described above, except that the lace holes **2426a-b** can be configured to direct the lace generally radially away from the base member **2414** rather than axially away from the base member **2214** as shown, for example, in FIG. **62**. Also, the lace holes **2426a-b** are placed generally on the same side of the base member **2414**, rather than on opposite ends as in the base member **2214** discussed above. Many variations are possible depending on the particular application to which the lacing system is applied. For example, in some embodiments, the base member can include only one lace hole and only one end of the lace can enter the housing and attach to the spool member. In these embodiments, the other end of the lace can attach to the base member or to the article being tightened.

(234) FIG. **86** is a cross sectional view of another embodiment of a knob core **2596** which can be used in a reel that can be similar in many ways to the reel **2204** described herein. The knob core **2596** can include pawls **2536** which can be integrally formed with the knob core **2596** to simplify construction and assembly of the reel. In other embodiments, the pawls **2536** can be attached to the knob core **2596** in any suitable manner. The pawls **2536** can include pawl arms **2532** which can be made of a material, thickness, and length so as to be flexible to allow the pawls **2536** to be displaced radially inwardly by housing teeth as the knob core **2596** is rotated in the tightening direction (shown by arrow A) in a manner similar to that described above. The pawls **2536** can include pawl teeth **2538a-b** formed at the ends of the pawl arms **2532**. In the illustrated embodiment two pawl teeth **2538a-b** are used per pawl **2536**, but any other suitable number of pawl teeth **2538a-b** can be used.

(235) When the knob core **2596** is twisted in the loosening direction (shown by arrow B), the pawl teeth **2538a-b** can bear against housing teeth (not shown in FIG. **86**) to prevent the knob core **2596** from rotating in the loosening direction. The force arrows drawn in FIG. **86** illustrate the directions in which the force is distributed radially. As the pawl teeth **2538a-b** bear against the housing teeth, a force is applied from the pawl teeth **2538a-b** to the housing teeth as shown. The pawl arms **2532** can be curved as shown so that, when the pawl teeth **2538a-b** bear against the housing teeth, the pawl arms **2532** tend to flex or buckle radially outwardly as shown by arrows in FIG. **86**. The pawls **2536** can be configured such that the housing teeth abut against the pawl arms **2532** such that, as the pawl arms **2532** attempt to flex or buckle radially outwardly, they bear against the tips of the housing teeth, distribute the force radially to the housing teeth, and are prevented from buckling. In some embodiments, the housing teeth can substantially prevent the pawl arms **2532**

from moving radially outwardly. Because pawls 2536 engage the housing teeth radially, not axially, and because the pawls 2536 are configured to be displaced radially, not axially, during tightening, substantially none of the force applied when twisting in the loosening direction is applied axially thereby reducing or eliminating the occurrence of unintentional axial movement of the knob core 2596 from the engaged position to the disengage position.

(236) Although the disclosure is discussed in terms of certain embodiments, it should be understood that the disclosure is not limited to the embodiments specifically shown and discussed. The embodiments are explained herein by way of example, and there are numerous modifications, variations, and other embodiments that may be employed within the scope of the present inventions. Components can be added, removed, and/or rearranged both with the individual embodiments discussed herein and between the various embodiments. For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. It should be understood that not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those of skill in the art will recognize that the inventions may be embodied or carried out in a manner that achieves one advantage or a group of advantages at taught or suggested herein without necessarily achieving other advantages as may be taught or suggested herein.

(237) Although various embodiments of lacing systems are described herein, the various components, features, or other aspects of the embodiments of the lacing systems described herein can be combined or interchanged to form additional embodiments of lacing systems not explicitly described herein, all of which are contemplated as being a part of the present disclosure. In addition, while a number of variations have been shown and described in detail, other modifications, which are within the scope of the this disclosure, will be readily apparent to those of skill in the art based upon this disclosure. Thus, it is intended that the scope of the disclosure should not be limited by the particular disclosed embodiments described above.

Claims

1. A reel based closure device for use in a lacing system, the reel based device comprising: a housing having a cylindrical wall; a plurality of teeth; a spool rotatably positioned within the housing, the spool being rotatable in a tightening direction to wind a tension member about the spool and thereby tighten the lacing system and the spool being rotatable in a loosening direction to unwind the tension member from about the spool and thereby loosen the lacing system; a knob supported by the housing, the knob being rotatable with respect to the housing, the knob being coupled to the spool such that a rotation of the knob in the tightening direction causes the spool to also rotate in the tightening direction; and a knob core that is separate from the knob and removably couplable thereto, the knob core including a cylindrical outer wall and one or more pawls, in which at least one of the one or more pawls includes a pawl beam and a pawl spring integrally formed with the pawl beam, the pawl beam being movable between a first position and a second position and the pawl spring being configured to bias the pawl beam toward the first position; wherein in the first position, a distal end of the pawl beam engages with the teeth to prevent the spool from rotating in the loosening direction when a loosening force is applied to the spool; wherein in the second position, the distal end of the pawl beam is displaced away from the teeth to allow the spool to rotate within the housing; wherein the knob core's cylindrical outer wall has a diameter that is larger than a diameter of the housing's cylindrical wall such that the knob core's cylindrical outer wall is positionable on an exterior of the housing's cylindrical wall; and wherein the one or more pawls are positioned about the knob core so that the one or more pawls are positioned on an interior of the housing's cylindrical wall.

2. The reel based closure device of claim 1, wherein the knob core also includes a cylindrical inner wall, and wherein the knob core's cylindrical inner wall has a diameter that is smaller than the

diameter of the housing's cylindrical wall such that the knob core's cylindrical inner wall is positionable on the interior of the housing's cylindrical wall.

3. The reel based closure device of claim 2, wherein the one or more pawls are positioned between the knob core's cylindrical inner wall and the knob core's cylindrical outer wall.
4. The reel based closure device of claim 2, wherein the knob core includes an annular channel that is defined between the knob core's cylindrical inner wall and the knob core's cylindrical outer wall.
5. The reel based closure device of claim 1, wherein the knob is axially moveable relative to the housing to move the one or more pawls out of engagement with the teeth and thereby allow the spool to rotate in an opposite direction.
6. The reel based closure device of claim 3, wherein the reel based closure device further comprises a bushing that is operably coupled with the knob to maintain the knob in an axially raised or an axially lowered position.
7. The reel based closure device of claim 1, wherein the one or more pawls are separate components from the knob core.
8. The reel based closure device of claim 1, wherein the one or more pawls are integrally connected or formed with the knob core.
9. The reel based closure device of claim 1, wherein the knob core includes a plurality of axially extending teeth that engage corresponding axially extending teeth of the spool.
10. A reel based closure device comprising: a housing having a cylindrical wall; a spool rotatably positioned within the housing, the spool being configured so that a tension member is windable about the spool; a knob supported by the housing and rotatable relative thereto, the knob being operably coupled with the spool such that a rotation of the knob causes the spool to rotate in a first direction within the housing to wind the tension member about the spool; and a knob core that is removably couplable with the knob, the knob core including a cylindrical outer wall and at least one pawl that is engageable with teeth to allow the spool to rotate in the first direction within the housing while preventing rotation of the spool in a second direction; wherein a distal end of the at least one pawl engages with the teeth to prevent the spool from rotating in the second direction when a loosening force is applied to the spool and the distal end of the at least one pawl is displaced away from the teeth as the spool is rotated in the first direction within the housing; wherein the knob core's cylindrical outer wall has a diameter that is larger than a diameter of the housing's cylindrical wall such that the knob core's cylindrical outer wall is positionable on an exterior of the housing's cylindrical wall; and wherein the at least one pawl is positioned about the knob core so that the at least one pawl is positioned on an interior of the housing's cylindrical wall.
11. The reel based closure device of claim 10, wherein the knob core also includes a cylindrical inner wall, and wherein the knob core's cylindrical inner wall has a diameter that is smaller than the diameter of the housing's cylindrical wall such that the knob core's cylindrical inner wall is positionable on the interior of the housing's cylindrical wall.
12. The reel based closure device of claim 11, wherein the at least one pawl is positioned between the knob core's cylindrical inner wall and the knob core's cylindrical outer wall.
13. The reel based closure device of claim 11, wherein the knob core includes an annular channel that is defined between the knob core's cylindrical inner wall and the knob core's cylindrical outer wall.
14. The reel based closure device of claim 10, wherein the knob is axially moveable relative to the housing to move the one or more pawls out of engagement with the teeth and thereby allow the spool to rotate in an opposite direction.
15. The reel based closure device of claim 14, wherein the reel based closure device further comprises a bushing that is operably coupled with the knob to maintain the knob in an axially raised or an axially lowered position.
16. The reel based closure device of claim 10, wherein the one or more pawls are separate components from the knob core.

17. The reel based closure device of claim 10, wherein the one or more pawls are integrally connected or formed with the knob core.

18. A reel based closure device comprising: a housing having a cylindrical wall; a spool rotatably positioned within the housing; a knob supported by the housing and rotatable relative thereto, the knob being operably coupled with the spool such that a rotation of the knob causes the spool to rotate within the housing; and a knob core having a cylindrical outer wall and at least one pawl that is engageable with teeth to allow the spool to rotate in a first direction within the housing while preventing rotation of the spool in a second direction; wherein the knob core's cylindrical outer wall has a diameter that is larger than a diameter of the housing's cylindrical wall such that the knob core's cylindrical outer wall is positionable on an exterior of the housing's cylindrical wall; and wherein the at least one pawl is positioned about the knob core so that the at least one pawl is positioned on an interior of the housing's cylindrical wall.

19. The reel based closure device of claim 18, wherein the knob core also includes a cylindrical inner wall, and wherein the knob core's cylindrical inner wall has a diameter that is smaller than the diameter of the housing's cylindrical wall such that the knob core's cylindrical inner wall is positionable on the interior of the housing's cylindrical wall.

20. The reel based closure device of claim 19, wherein the at least one pawl is positioned between the knob core's cylindrical inner wall and the knob core's cylindrical outer wall.

21. The reel based closure device of claim 19, wherein the knob core includes an annular channel that is defined between the knob core's cylindrical inner wall and the knob core's cylindrical outer wall.
